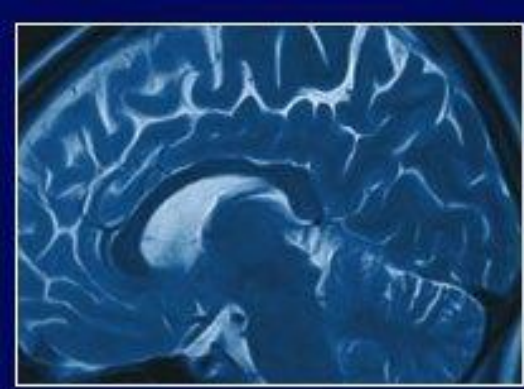




*The Waisman Laboratory
for Brain Imaging and Behavior*



University of Wisconsin
**SCHOOL OF MEDICINE
AND PUBLIC HEALTH**

Topology for Image Analysis

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Unit Objectives

Understand Euler characteristics (EC).

Understand Betti numbers

Understand how to compute EC

Spectral
geometry

Trees

Segmentation

Tubular
structures

Clustering

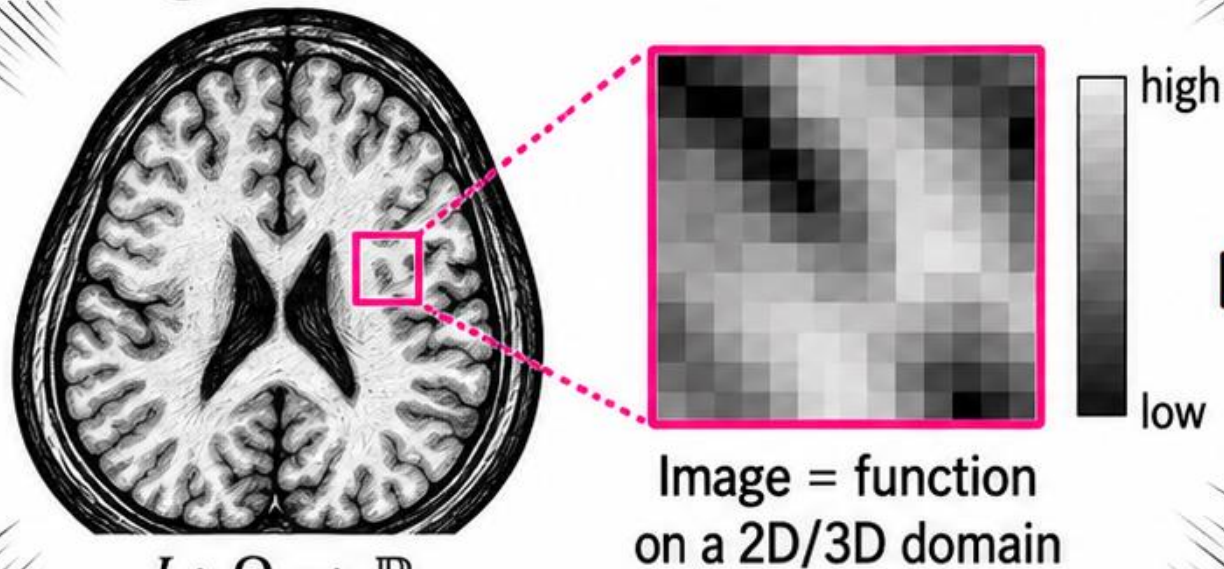
TDA

Persistent
homology

Topological
constraints

Topology
correction

Images as functions



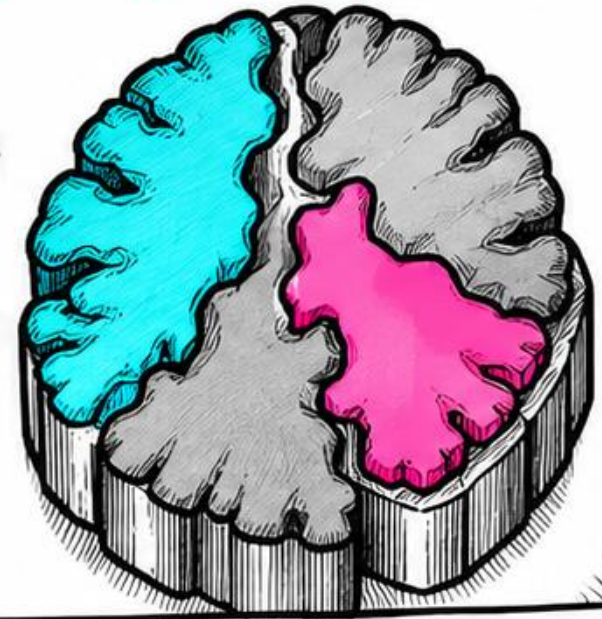
$$I : \Omega \rightarrow \mathbb{R}$$

(voxel intensity)

Segmentation as a topological operation

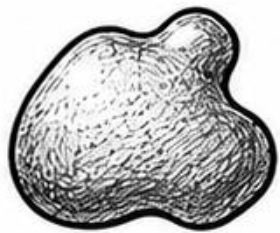
Partition Ω into disjoint regions $\{R_1, \dots, R_K\}$

$$\Omega = \bigcup_{k=1}^K R_k, \quad R_i \cap R_j = \emptyset \quad (i \neq j)$$

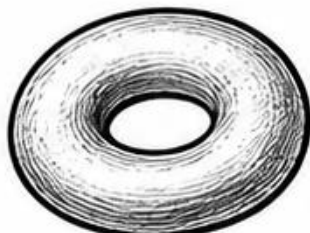


Why topology matters?

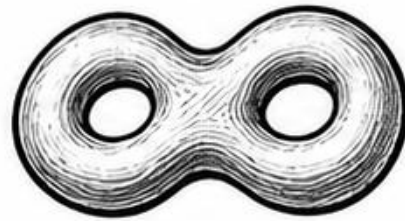
Images contain connected structures such as components, holes, and loops.



Component
(H_0)



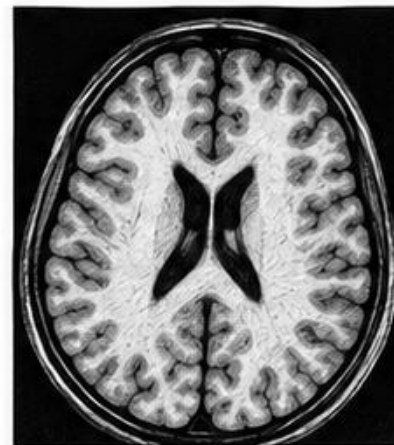
Hole
(H_1)



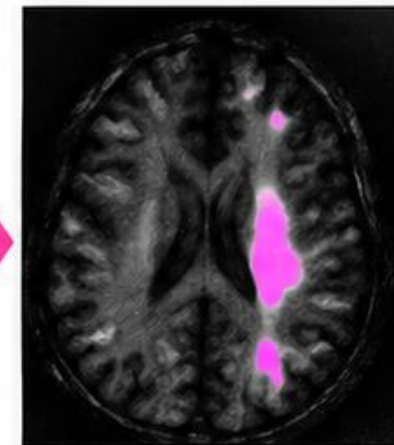
Two holes
($H_1 = 2$)

Segmentation should respect this topological structure, not just local intensity.

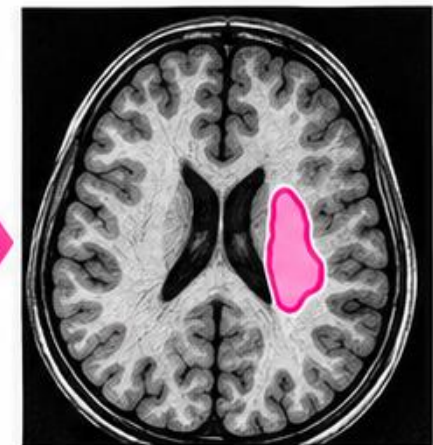
Example: stroke lesion segmentation



Original image



Lesion probabilities



Segmented regions

Topology-aware segmentation preserves coherent lesion regions, avoids spurious fragments, and provides a simpler, more meaningful representation of the image.

Stroke patients with dysphagia

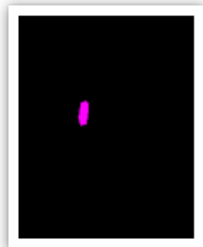


1dwi0034.bmp

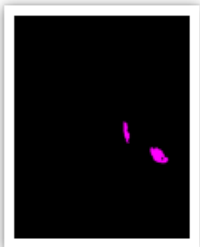
3dwi0033.bmp

5dwi0034.bmp

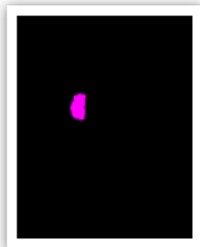
6dwi0034.bmp



7dwi0033.bmp



8dwi0034.bmp



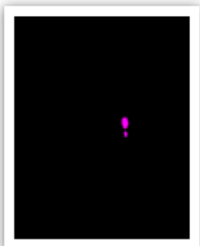
9dwi0034.bmp



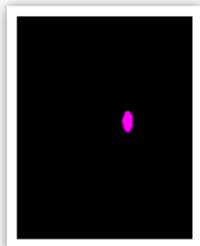
10dwi0035.bmp



17dwi0033.bmp



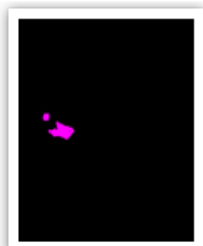
18dwi0035.bmp



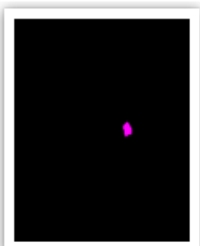
20dwi0034.bmp



22dwi0034.bmp



23dwi0034.bmp



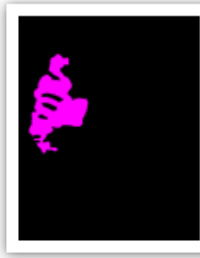
24dwi0034.bmp



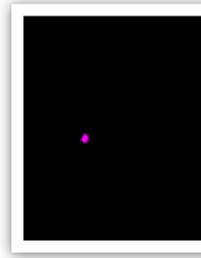
27dwi0033.bmp



28dwi0035.bmp



96dwi0034.bmp



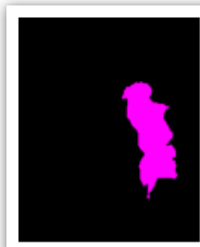
97dwi0034.bmp



98dwi0035.bmp



101dwi0034.bmp



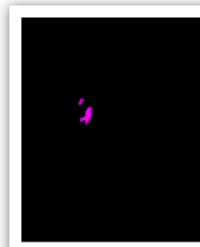
103dwi0034.bmp



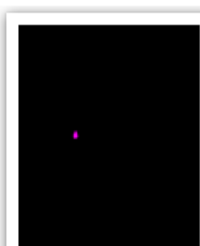
104dwi0036.bmp



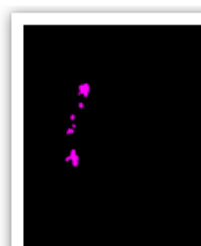
105dwi0035.bmp



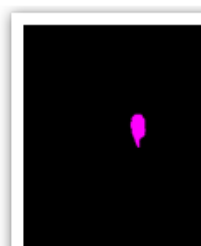
106dwi0033.bmp



107dwi0034.bmp



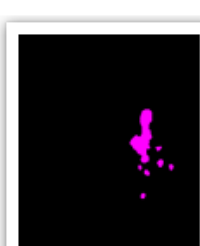
109dwi0035.bmp



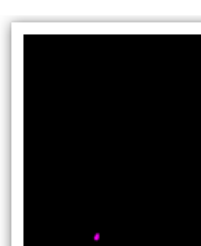
110dwi0033.bmp



111dwi0035.bmp



113dwi0072.bmp



116dwi0071.bmp



118dwi0073.bmp



120dwi0072.bmp

Group 0: patients who got improved after one month (n1=58)

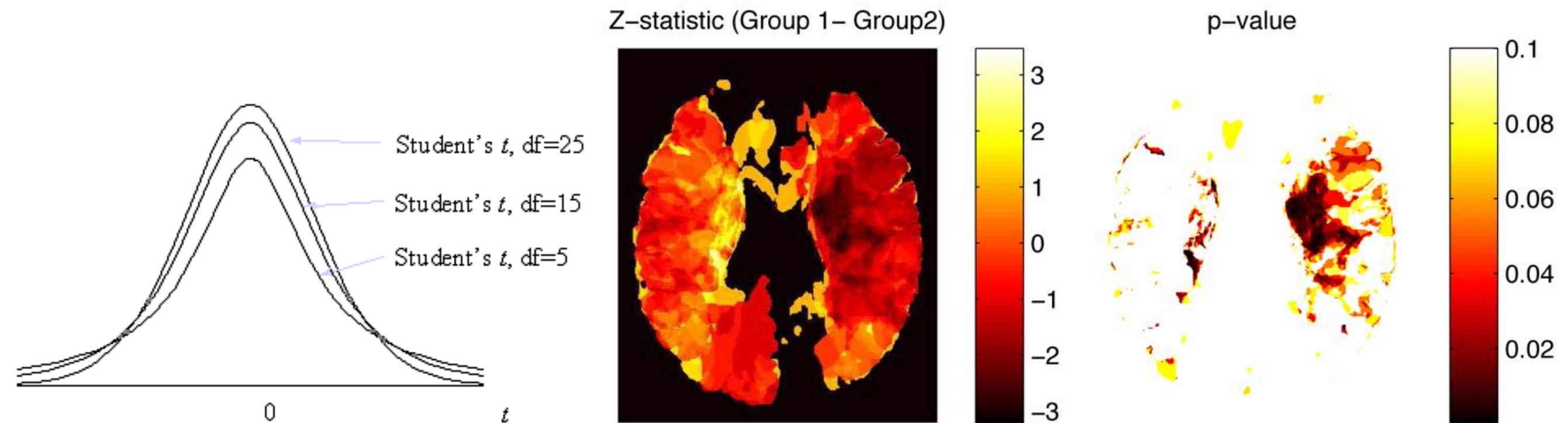
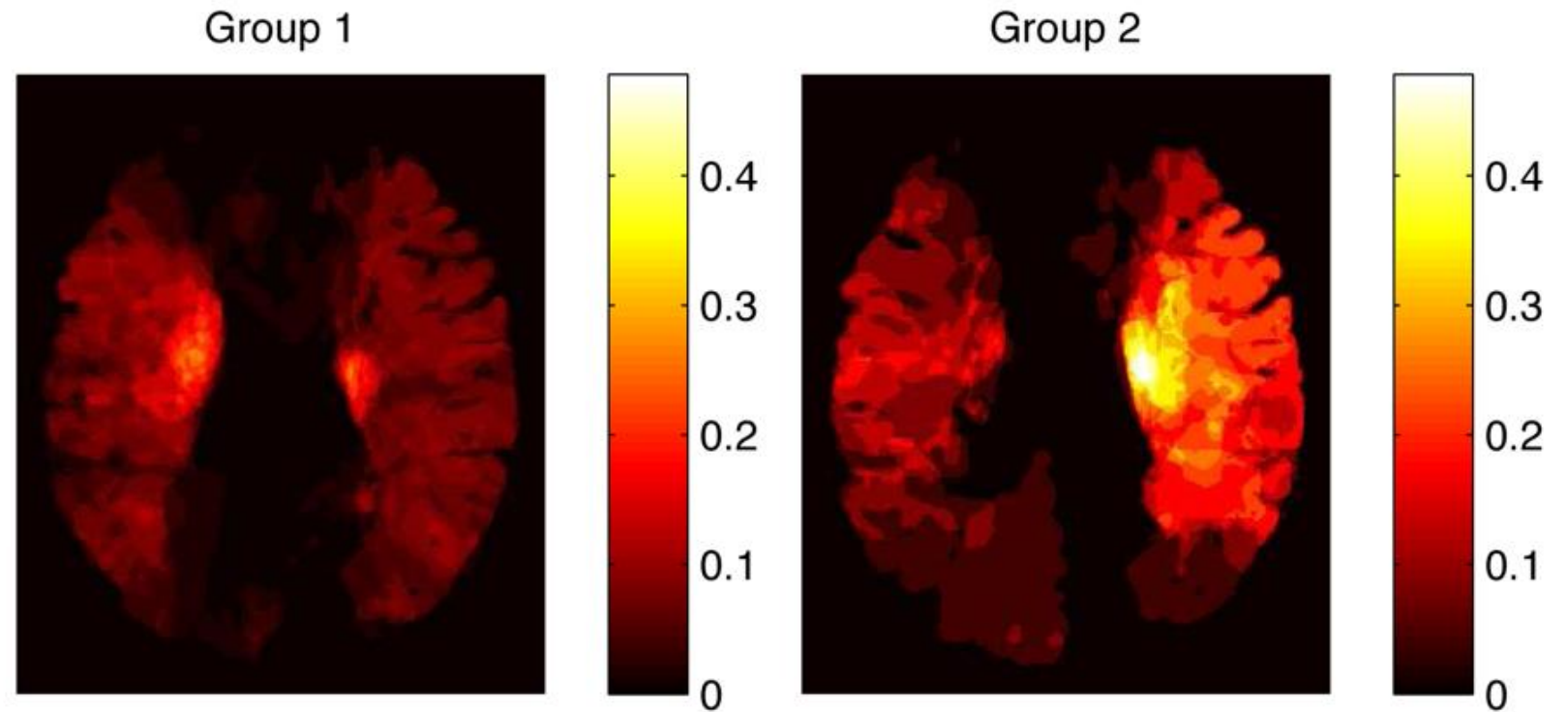
Group 1: patients who didn't get better (n2=23)

Question: How we discriminate the groups?

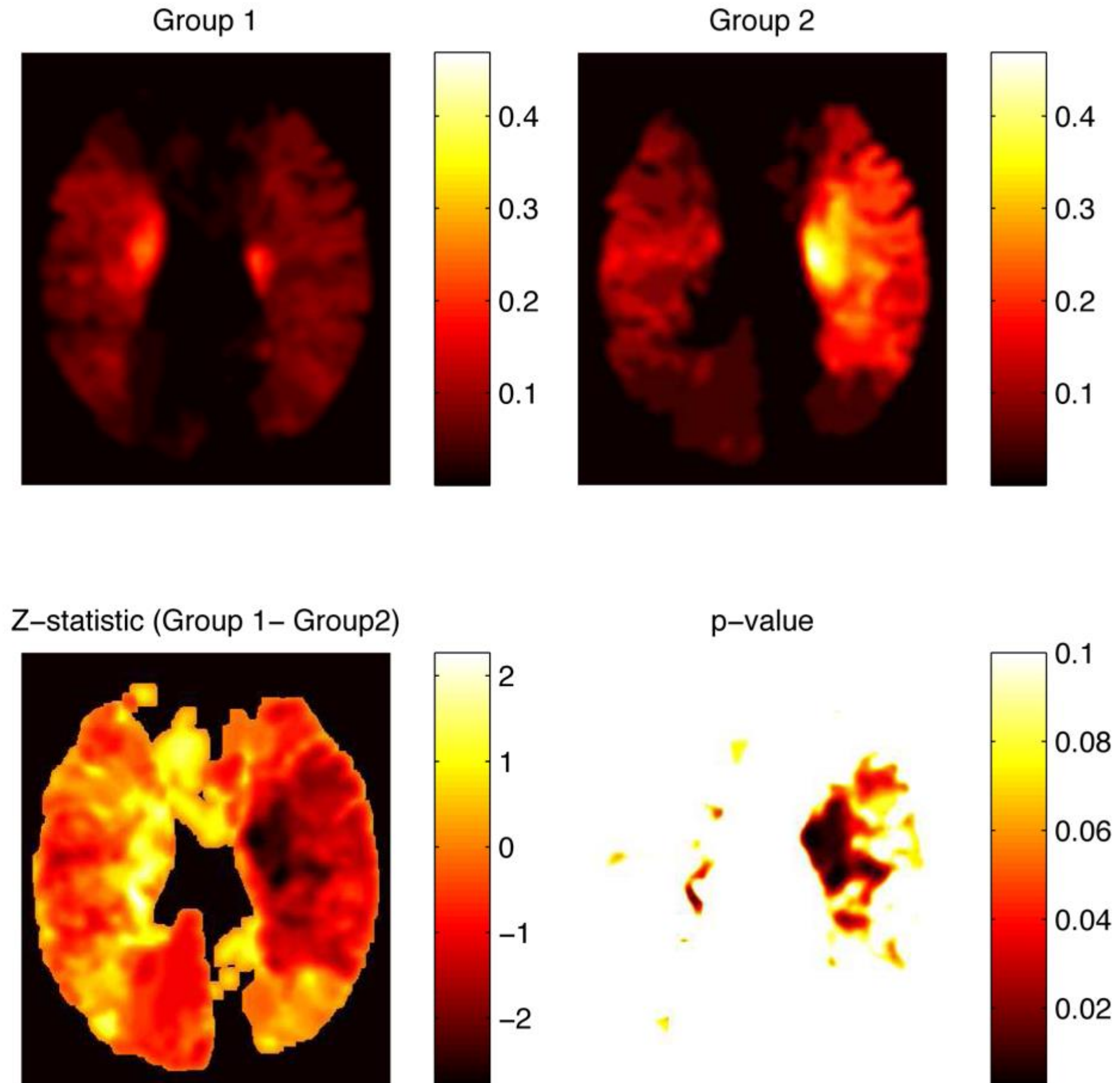
Two sample test at each pixel without smoothing

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{n_1+n_2-2}$$

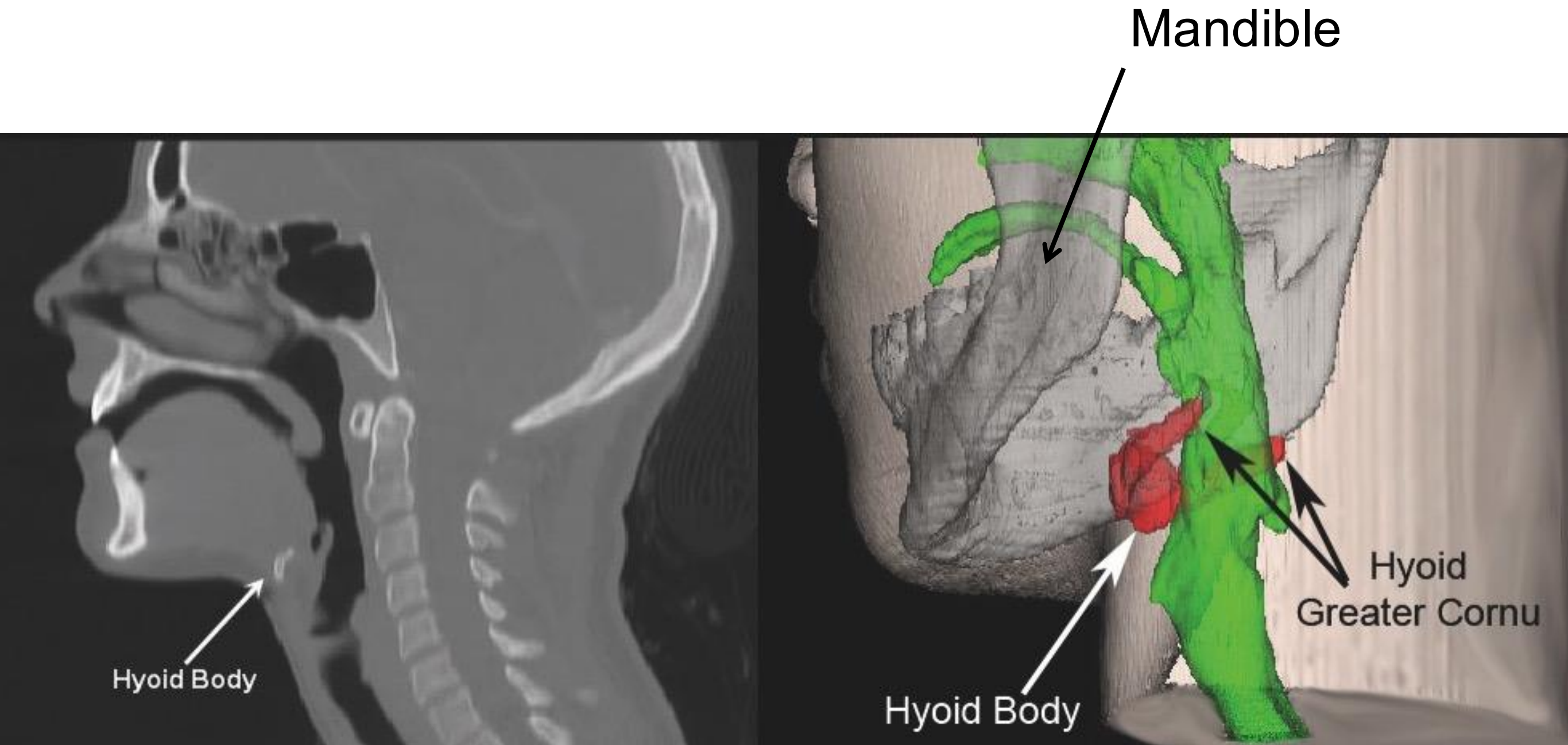
$$s^2 = \frac{\sum_{i=1}^{n_1} (x_i - \bar{x}_1)^2 + \sum_{j=1}^{n_2} (x_j - \bar{x}_2)^2}{n_1 + n_2 - 2}$$



Two sample test with smoothing

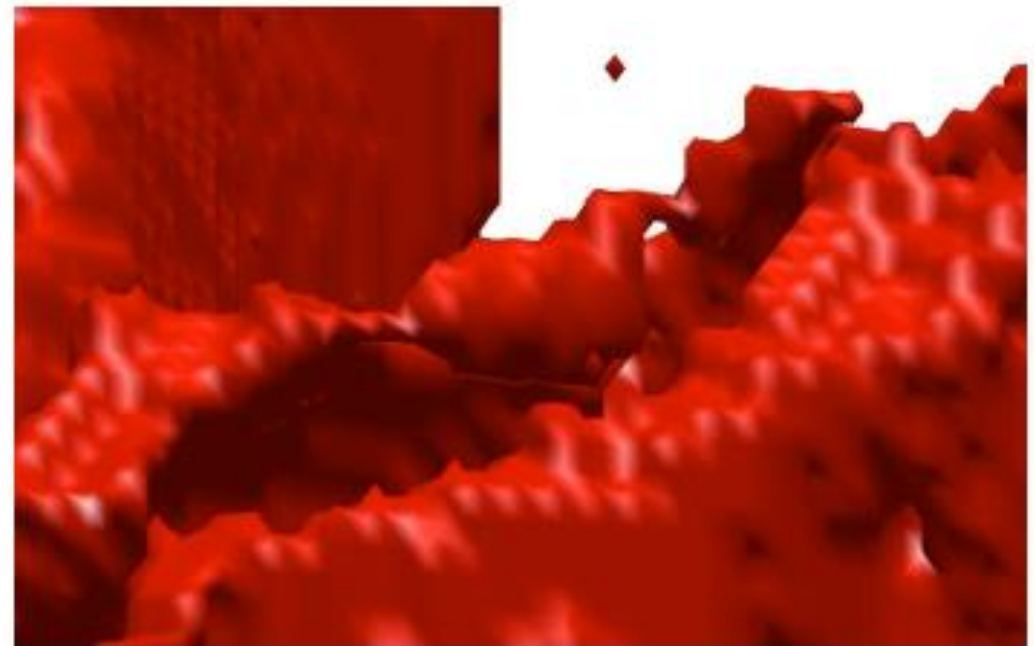
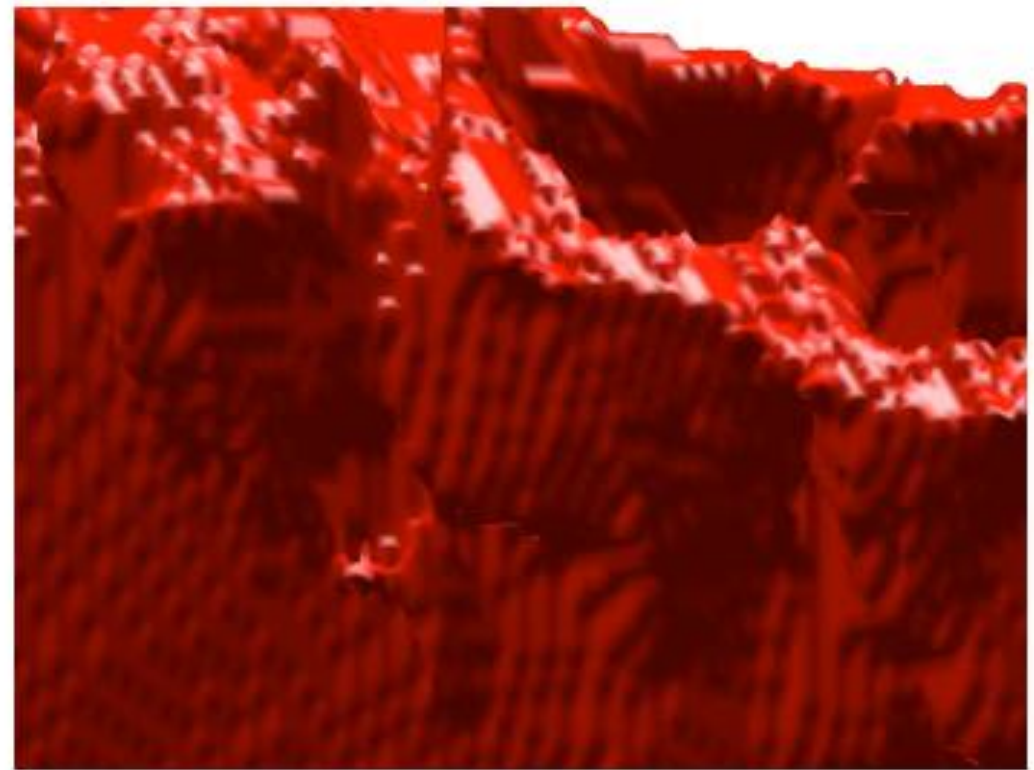
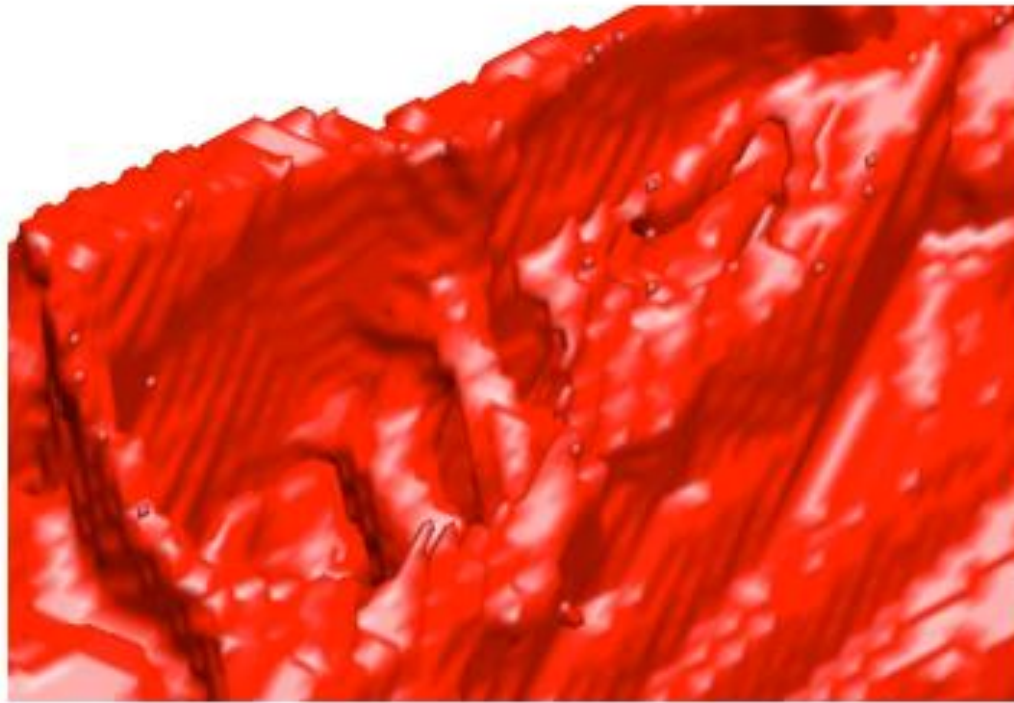


Head and neck CT image



Semiautomatic histogram thresholding in ANALYZE package

→ Automatic pipeline using **ANTS**-based template matching



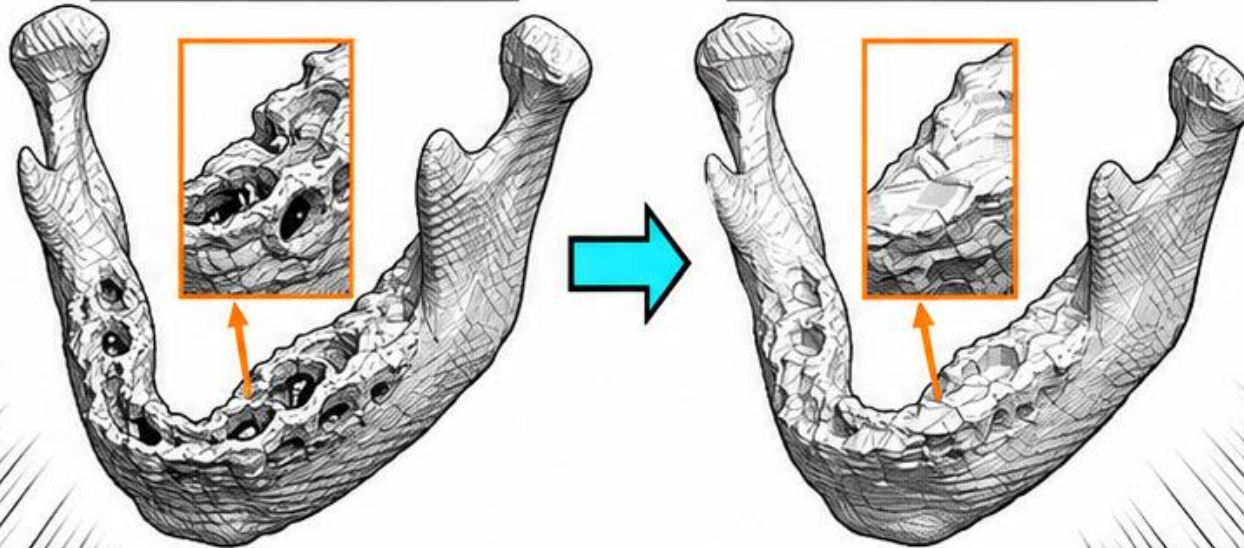
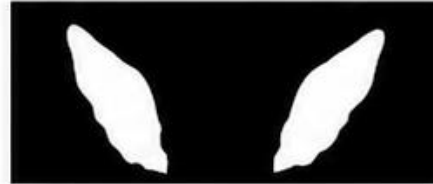
Volume rendering of a mandible from CT
Holes and handles in binary volume

1. Segmentation may create **topological errors**

Initial segmentation

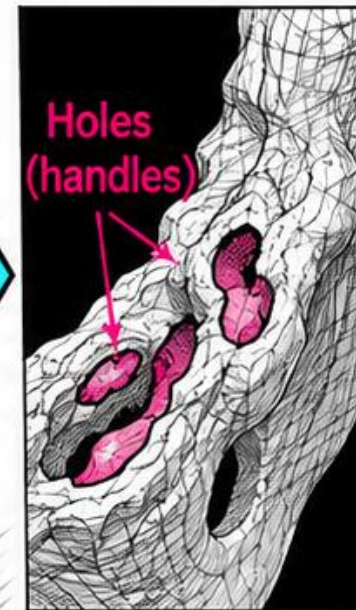


After naive correction

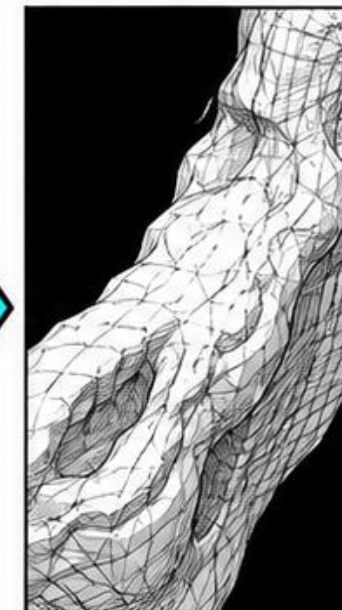


2. **Topology-aware** correction

Incorrect topology



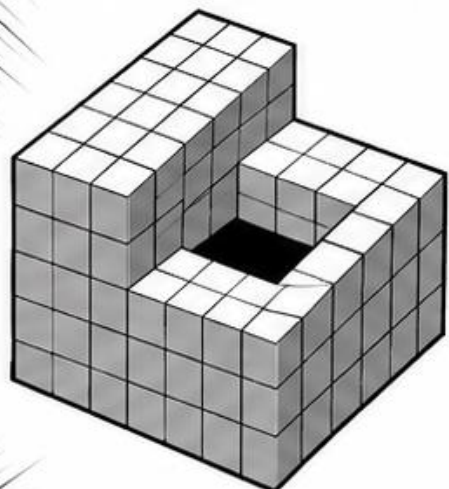
Topology-corrected



Preserve the correct topology using the Euler characteristic

$$\chi = \underbrace{\beta_0}_{\text{components}} - \underbrace{\beta_1}_{\text{holes}} + \underbrace{\beta_2}_{\text{voids}}$$

3. Compute **Euler characteristic** from the mask



Binary mask M

From the segmented mask, compute Betti numbers and Euler characteristic

$$\chi(M) = \beta_0 - \beta_1 + \beta_2$$

β_0 : number of connected components

β_1 : number of holes (handles)

β_2 : number of enclosed voids

4. How to check **without seeing images?**

- ✓ Compute χ for the segmented mask.
- ✓ Compare with the expected topology (e.g., one component, no holes: $\chi = 1$).
- ✓ Flag a topological defect if χ is incorrect (e.g., $\chi \neq 1$ or $\beta_1 > 0$).

Example: mandible

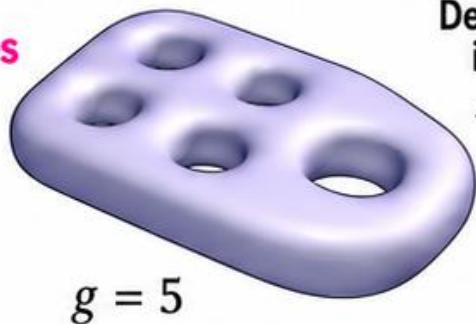
Mask	χ
Initial result	-1
Topology-corrected	1
Expected	1

Correct topology confirmed numerically.

k-cyles

1. Genus: number of holes

Genus g is the number of **holes** in a surface or an object.



Deformation invariant



Objects can be continuously deformed without changing the number of holes (genus).

2. Euler characteristic (χ)

Most widely used topological invariant.

$$\chi = N - E + F - V$$

N : # vertices (0-cells)
 E : # edges (1-cells)
 F : # faces (2-cells)
 V : # volumes (3-cells)

2D surface (no volume)

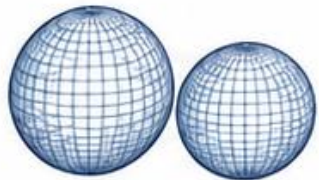
$$\chi = N - E + F$$

3D object

$$\chi = N - E + F - V$$

3. Examples: Euler characteristic and genus

Two spheres (disconnected)



$$g = 0$$

$$\chi = 4$$

Sphere



$$g = 0$$

$$\chi = 2$$

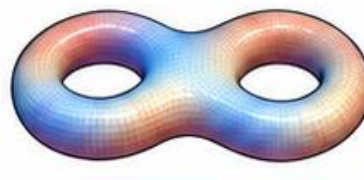
Torus (1 handle)



$$g = 1$$

$$\chi = 0$$

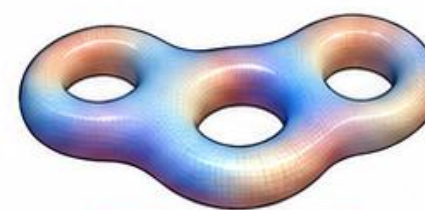
Double torus (2 handles)



$$g = 2$$

$$\chi = -2$$

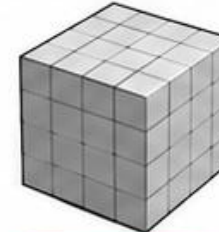
Three-handle surface



$$g = 3$$

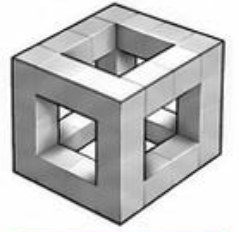
$$\chi = -4$$

Solid cube (3D, no hole)



$$\chi = 1$$

Cubic shell (3D, one hole)



$$\chi = 0$$

4. Why $\chi = 2 - 2n$ for n -handles? (2D closed surface)

Start with a sphere ($g = 0, \chi = 2$)



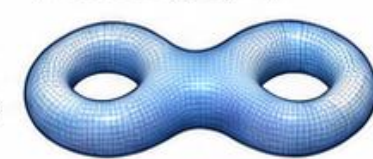
Add one handle (reduces χ by 2)



$$g = 1$$

$$\chi = 0$$

Add another handle (reduces χ by 2)



$$g = 2$$

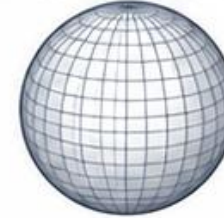
$$\chi = -2$$

...

After adding n handles: $\chi = 2 - 2n$

5. Sketch of proof (closed orientable surface)

A closed orientable surface with genus n is topologically equivalent to a sphere with n handles.



$\longleftrightarrow$



n handles

Thus, the Euler characteristic depends only on the genus and is a deformation-invariant topological property.

Key idea:

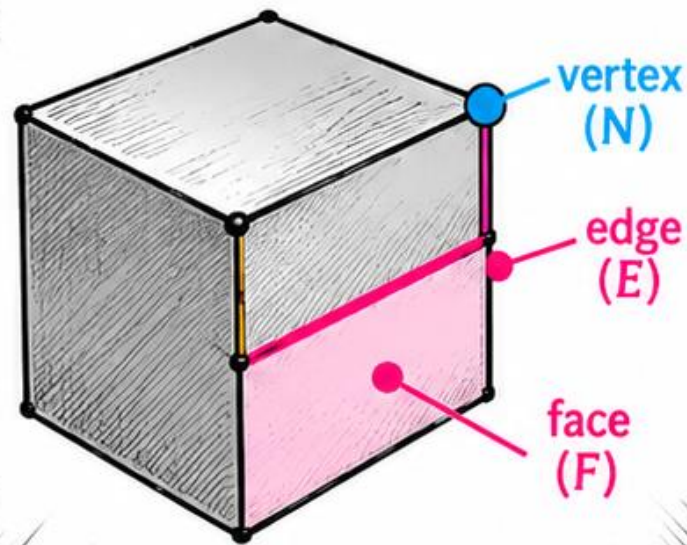
- Start with a sphere ($\chi = 2$).
- Each handle (torus) reduces χ by 2.
- Therefore, for n handles, $\chi = 2 - 2n$.

Relation:

$$EC = 2 - 2g$$

1. Polyhedron

A polyhedron is a solid object with polygonal faces, edges and vertices.



2. Euler characteristic

$$EC = N - E + F - V$$

N : number of vertices (0-cells)

E : number of edges (1-cells)

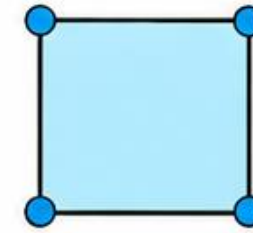
F : number of faces (2-cells)

V : number of volumes (3-cells)

For 2D objects (no volume):

$$EC = N - E + F$$

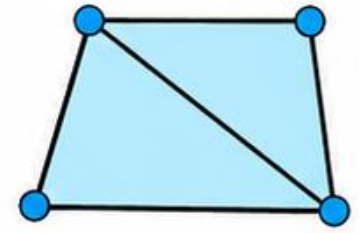
3. Examples in 2D



$$\begin{aligned} EC &= N - E + F \\ &= 4 - 4 + 1 \\ &= 1 \end{aligned}$$

If there is no face

$$\begin{aligned} EC &= N - E \\ &= 4 - 4 \\ &= 0 \end{aligned}$$

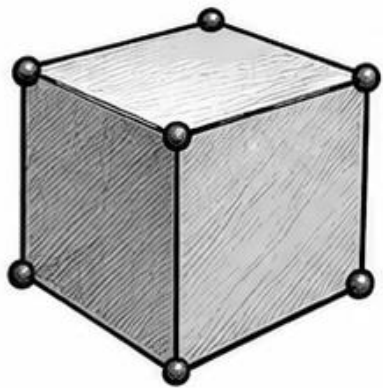


$$\begin{aligned} EC &= N - E + F \\ &= 4 - 6 + 4 \\ &= 2 \end{aligned}$$

If there is no volume

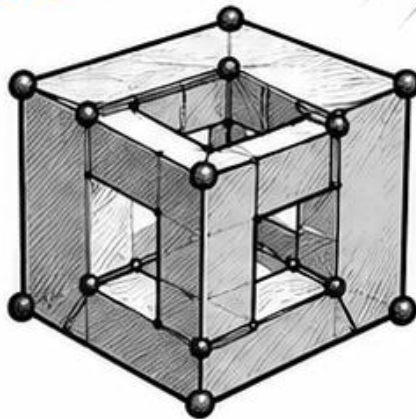
$$\begin{aligned} EC &= N - E + F \\ &= 4 - 6 + 4 \\ &= 2 \end{aligned}$$

4. Examples in 3D



Solid cube (no hole)

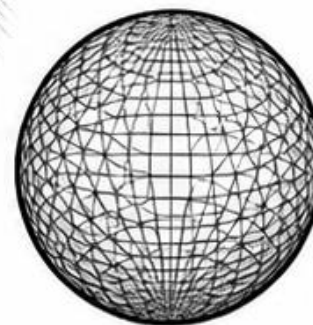
$$\begin{aligned} EC &= N - E + F - V \\ &= 8 - 12 + 6 - 1 \\ &= 1 \end{aligned}$$



Cubic shell (one hole)

$$\begin{aligned} EC &= N - E + F - V \\ &= 16 - 32 + 24 - 8 \\ &= 0 \end{aligned}$$

5. Topological equivalence



Sphere

$$EC = 2$$



Polyhedral sphere

$$EC = 2$$

Topologically equivalent objects have the same Euler characteristic (deformation-invariant).

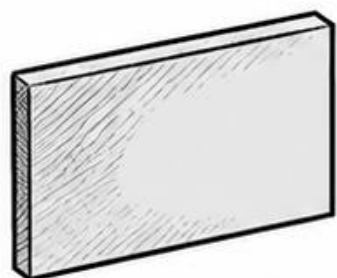
6. Interpretation

$EC \approx \text{\#dimension} - \text{\#holes}$

- Disk (2D, no hole):
 $EC = 2 - 0 = 2$
- Sphere (2D, one hole):
 $EC = 2 - 1 = 1$
- Solid ball (3D, no hole):
 $EC = 3 - 1 = 1$
- Torus (3D, one hole):
 $EC = 3 - 1 = 2 - 1 = 0$

EC is an approximate measure of data dimensionality!

1. Computing Euler characteristic by parts: simple examples

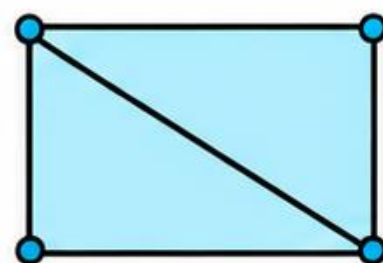


Cover an object with polyhedra (vertices, edges, faces).



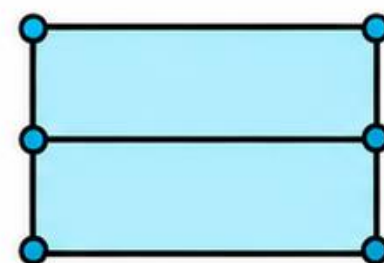
$$\begin{aligned} EC &= N - E + F \\ &= 4 - 4 + 1 \\ &= 1 \end{aligned}$$

(disk)



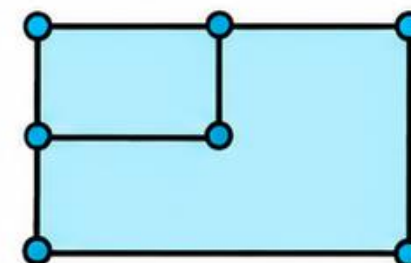
$$\begin{aligned} EC &= 4 - 5 + 2 \\ &= 1 \end{aligned}$$

(disk)



$$\begin{aligned} EC &= 6 - 7 + 2 \\ &= 1 \end{aligned}$$

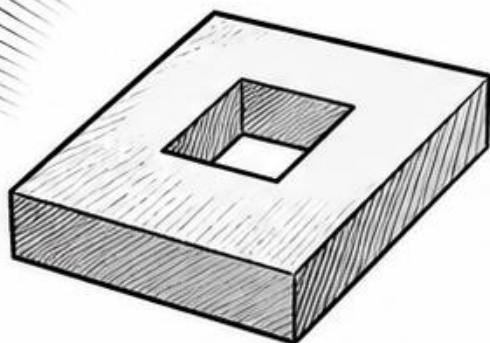
(disk)



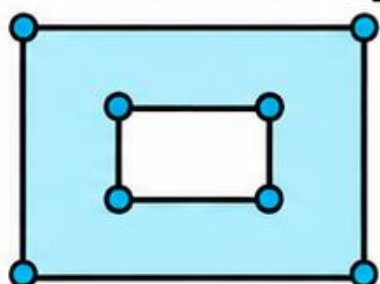
$$\begin{aligned} EC &= 7 - 8 + 2 \\ &= 1 \end{aligned}$$

(disk)

2. Objects with a hole: count by parts

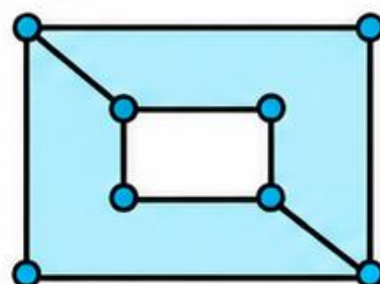


Cover an object with polyhedra.

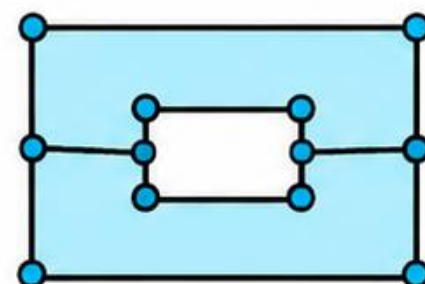


$$\begin{aligned} EC &= N - E + F \\ &= 8 - 8 + 1 \\ &= 1 \end{aligned}$$

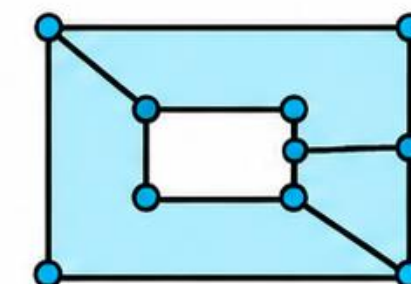
Incorrect computation
(misses the hole)



$$\begin{aligned} EC &= 8 - (8+2) + (1+1) \\ &= 0 \end{aligned}$$



$$\begin{aligned} EC &= (8+4) - (8+6) + (1+1) \\ &= 0 \end{aligned}$$



$$\begin{aligned} EC &= (8+2) - (8+4) + (1+1) \\ &= 0 \end{aligned}$$

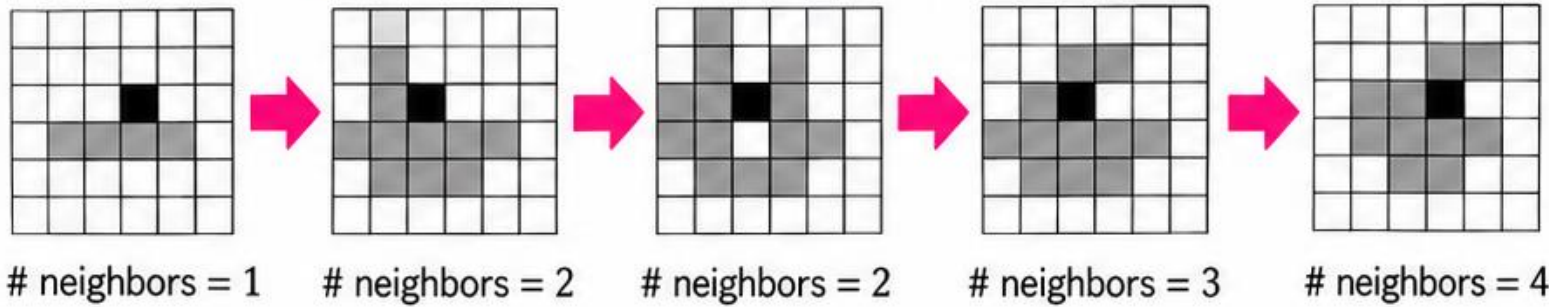
Correct computation
(detects the hole)

Euler characteristic: $\chi = N - E + F$

N : number of vertices (0-cells)
E : number of edges (1-cells)
F : number of faces (2-cells)

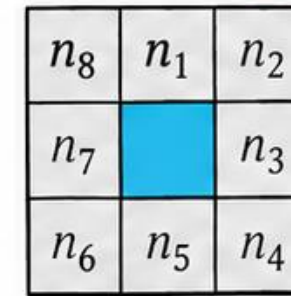
Disk: $\chi = 1$
One hole (annulus): $\chi = 0$

1. Online computation of Euler characteristic in 2D



# Vertex	2	1	0	0	0
# Edge	3	2	2	1	0
# Face	1	1	1	1	1
EC change	0	0	-1	0	1

2. Local update rules (2D)

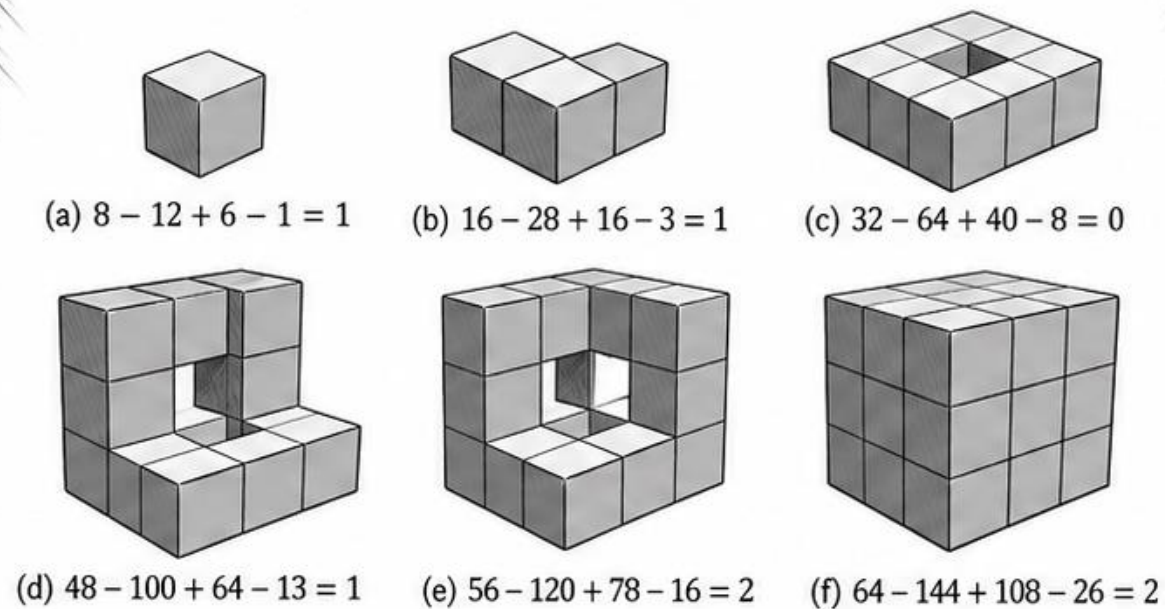


Let k be the number of existing neighbors ($n_1, \dots, n_8$) when adding a new pixel.

k	1	2	3	4
$\Delta\chi$	0	-1	0	1

Update $\chi \leftarrow \chi + \Delta\chi$ for each added pixel.

3. Computing Euler characteristic in 3D



In 3D: $\chi = \# \text{ vertices} - \# \text{ edges} + \# \text{ faces} - \# \text{ volumes}$

4. Iterative algorithm for a binary image

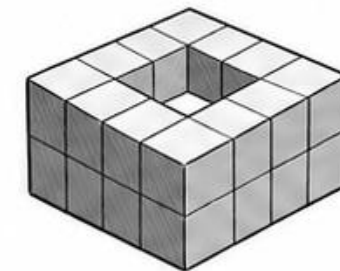
```

Input : binary image  $I$ 
Output: Euler characteristic  $\chi$ 
 $\chi \leftarrow 0$ 
for each foreground voxel  $p$ 
  (in any order):
     $k \leftarrow \#$  of existing foreground neighbors
     $\Delta\chi \leftarrow \text{lookup}(k)$ 
     $\chi \leftarrow \chi + \Delta\chi$ 
end
return  $\chi$ 

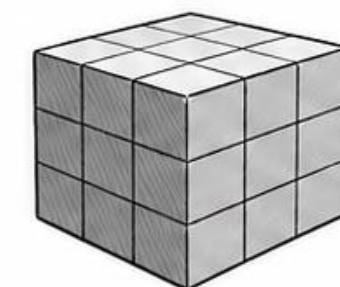
```

Use 4/8-neighbors in 2D or 6/26-neighbors in 3D.

5. Result and interpretation



$\chi = 0$
(one hole)



$\chi = 1$
(no hole)

Euler characteristic detects topological structure without explicit visualization.

Problem: Given a 2D binary image as input, code an iterative algorithm.

Expected Euler characteristic/Betti numbers

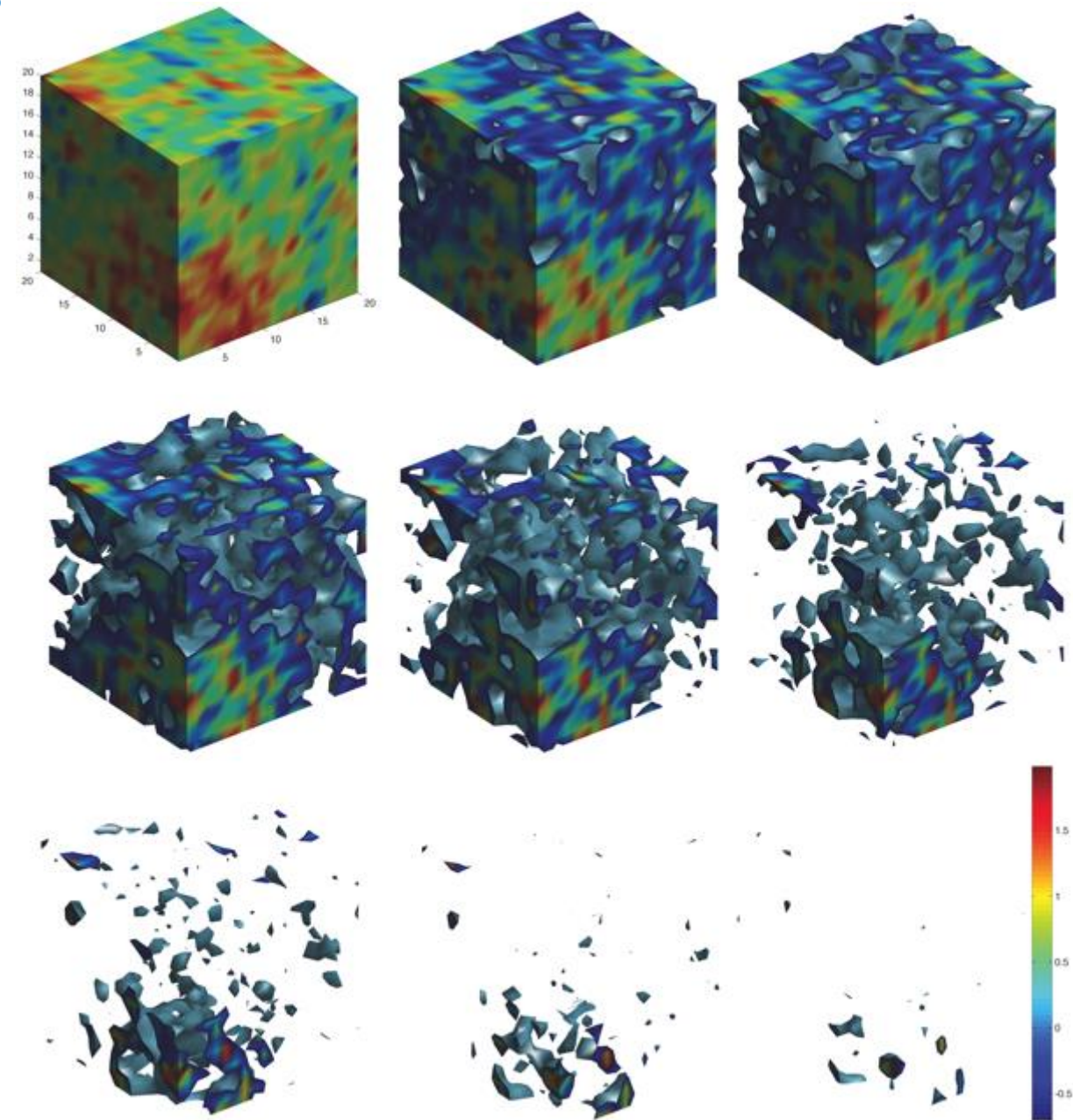
Random field, stochastic process

$$P\left(\sup_{x \in \mathbb{M}} T(x) > h\right)$$

$$A_h = \{x \in \mathbb{M} : T(x) > h\}$$

$$P\left(\sup_{x \in \mathbb{M}} T(x) > h\right) = \mathbb{E}\chi(A_h)$$

$$\chi(A_h) = \sum_j (-1)^j \beta_j(A_h)$$

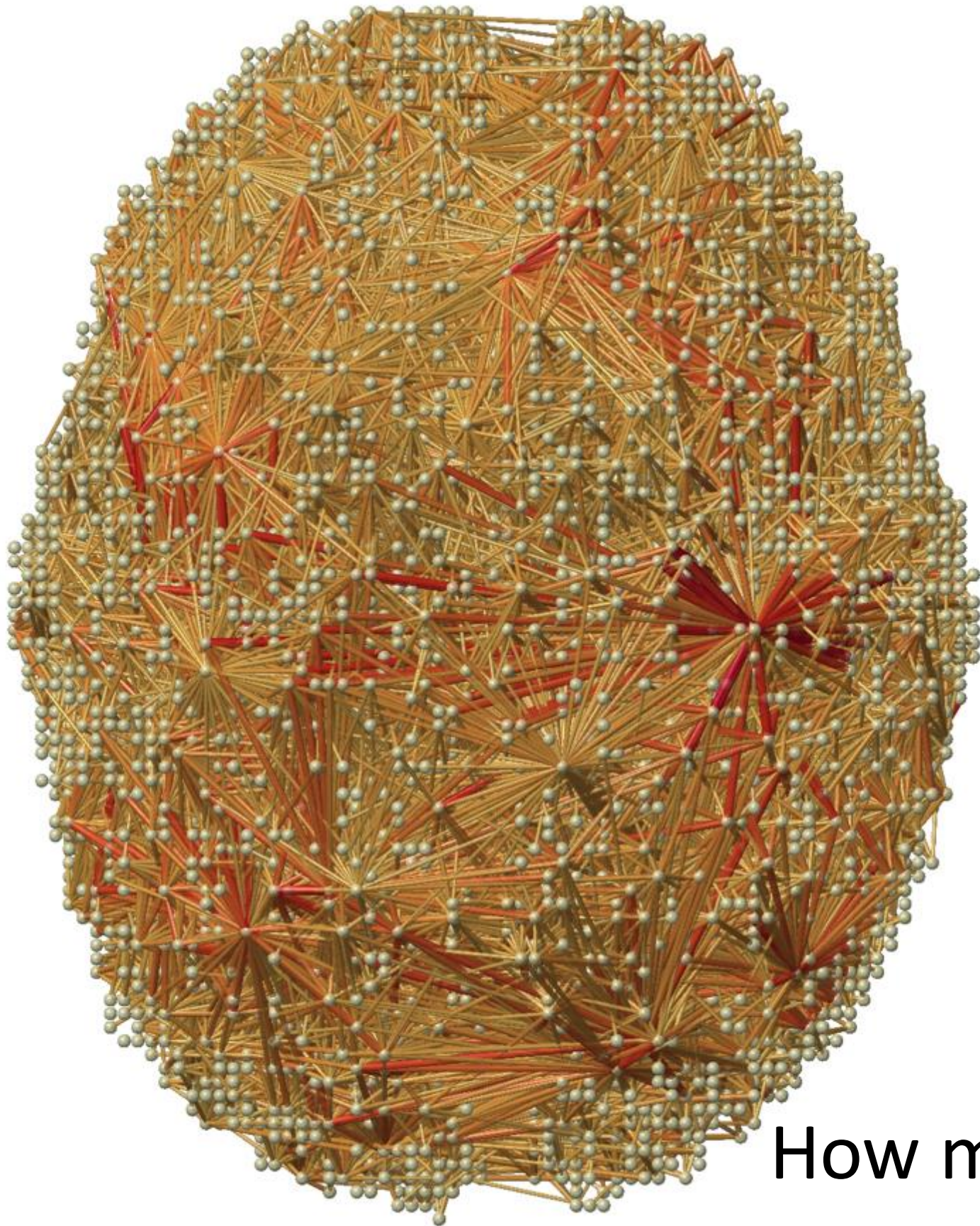


Milnor 1963 Morse theory

Adler, 1994 The geometry of random fields

Worsley et al., 1996 Human Brain Mapping

How to compute the number of cycles in big network data?

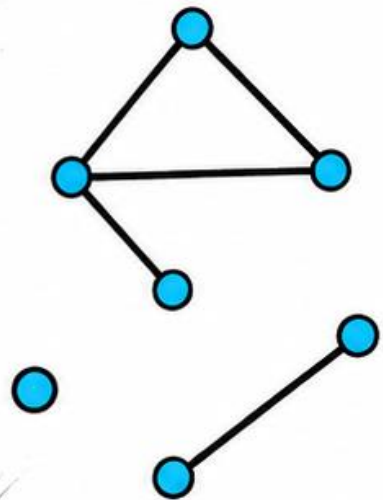


Radeon Vega64 eGPU

How many cycles in the network?

1. Betti numbers β_i

Count i -dimensional holes in a topological space.



β_0 = # of connected components
= **3**

β_1 = # of 1D loops
= **1**

Euler characteristic:

$$\chi = \beta_0 - \beta_1 = 3 - 1 = \mathbf{2}$$

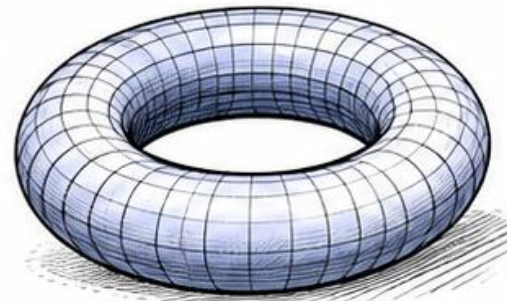
β_0 : connected components

β_1 : 1D holes (loops)

β_2 : 2D cavities (voids)

...

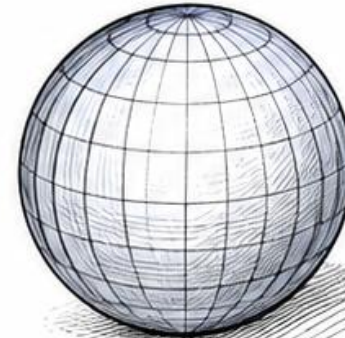
2. Betti numbers in higher dimensions



Torus

$$(\beta_0, \beta_1, \beta_2) = (1, 2, 0)$$

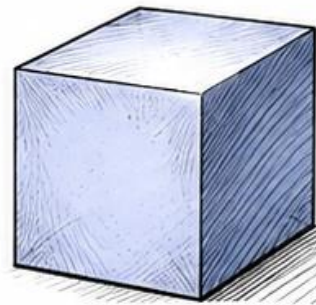
$$\chi = 1 - 2 + 0 = \mathbf{-1}$$



Sphere

$$(1, 0, 1)$$

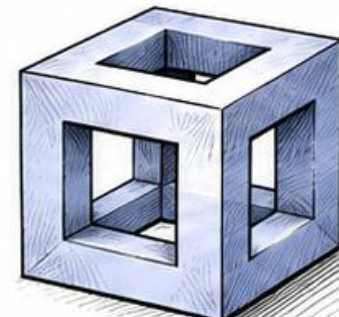
$$\chi = 1 - 0 + 1 = \mathbf{2}$$



Solid cube

$$(1, 0, 0)$$

$$\chi = 1 - 0 + 0 = \mathbf{1}$$



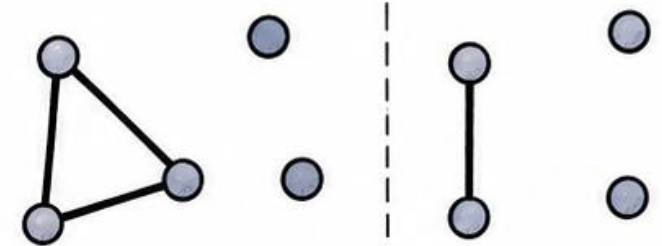
Cubic shell (one hole)

$$(1, 0, 1)$$

$$\chi = 1 - 0 + 1 = \mathbf{2}$$

$$\chi = \beta_0 - \beta_1 + \beta_2 - \dots$$

3. Betti numbers in graphs and networks



$$\beta_0 = \mathbf{2}$$

$$\beta_1 = \mathbf{1}$$

$$\beta_0 = \mathbf{3}$$

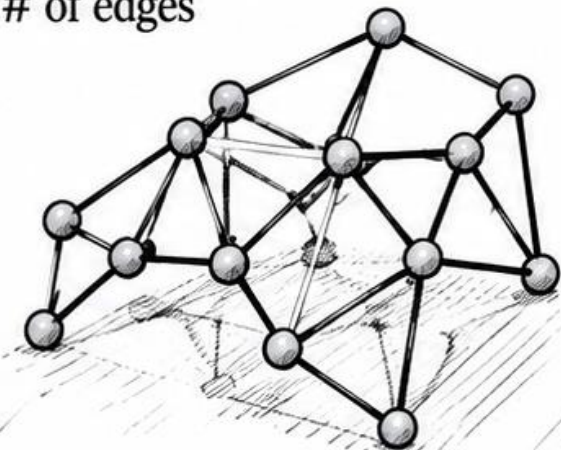
$$\beta_1 = \mathbf{0}$$

For a graph:

$$\chi = p - q = \beta_0 - \beta_1$$

p = # of vertices

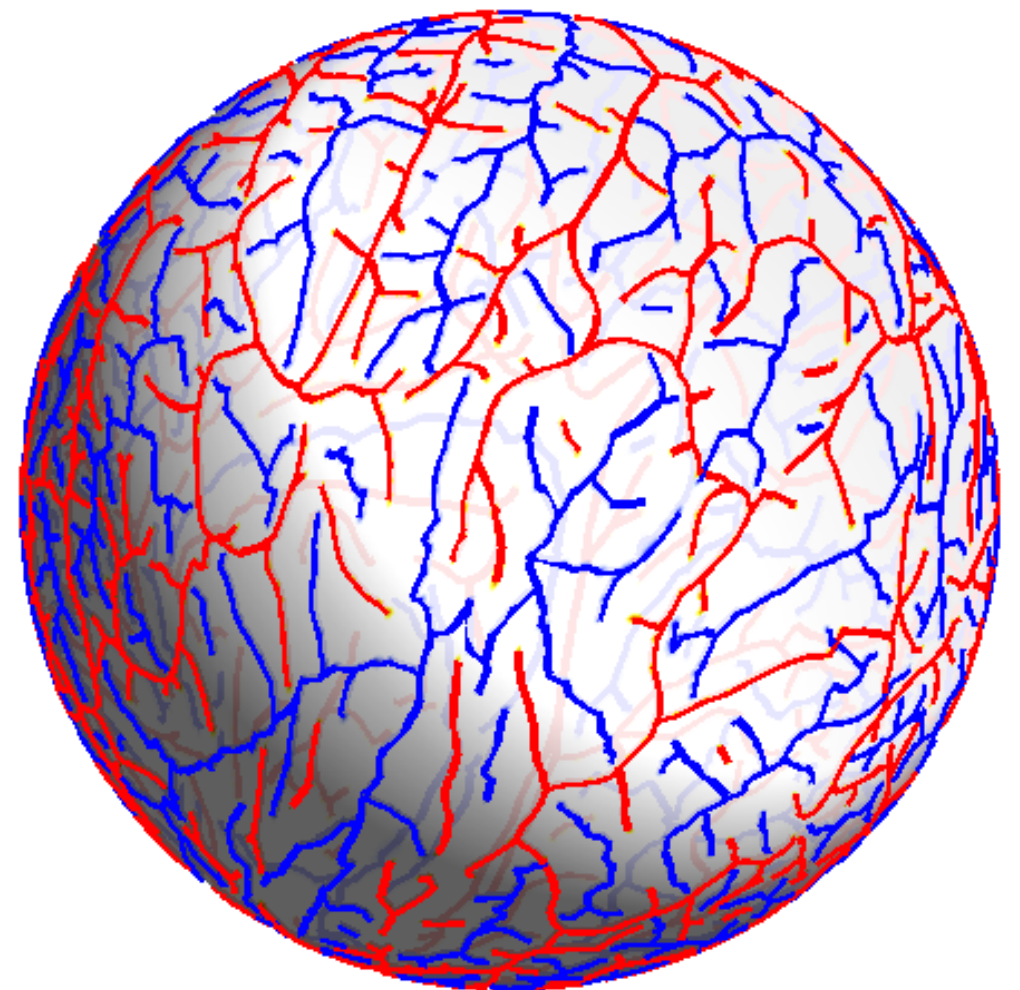
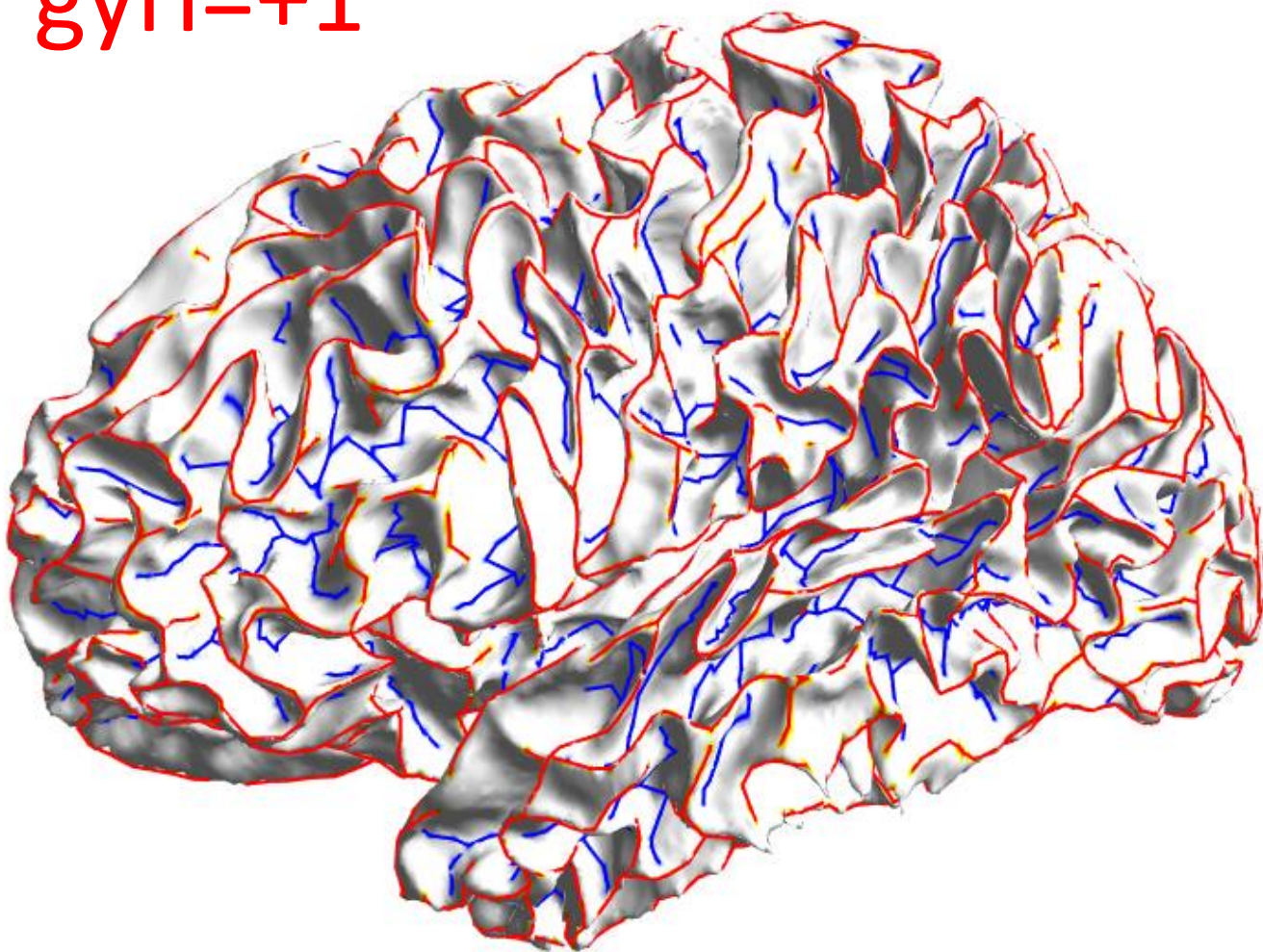
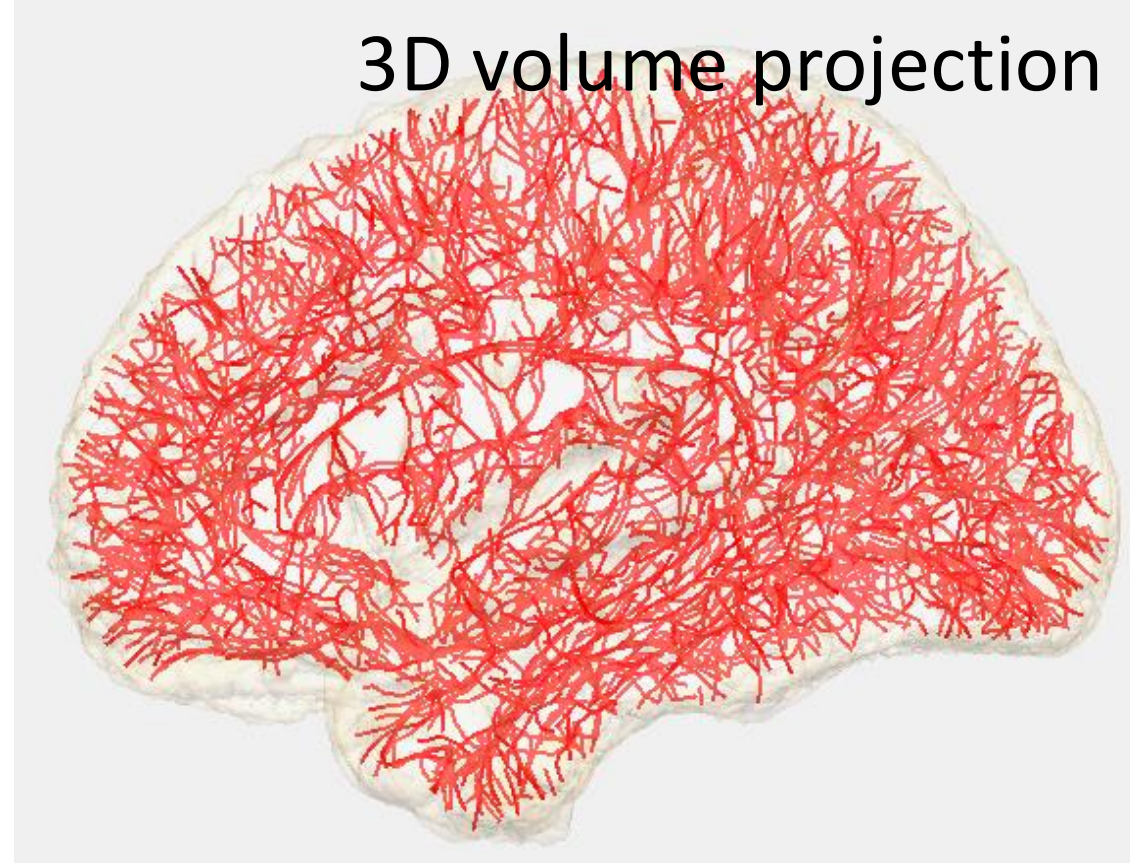
q = # of edges



Sulcal and gyral trees of brain from MRI

sulci = -1
gyri = +1

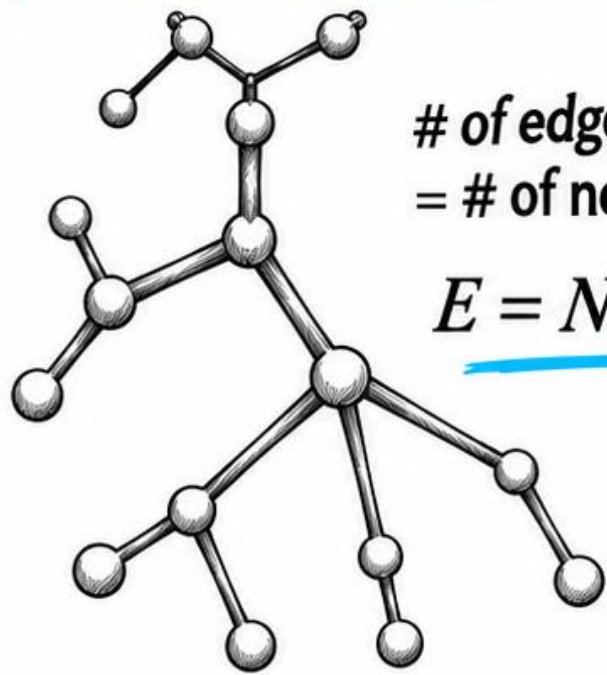
3D volume projection



White matter surface

Trees on manifold

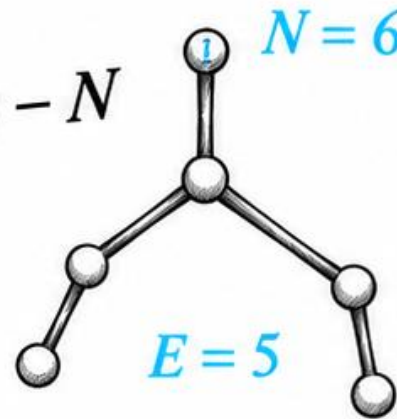
Topology of tree



of edges
= # of nodes - 1

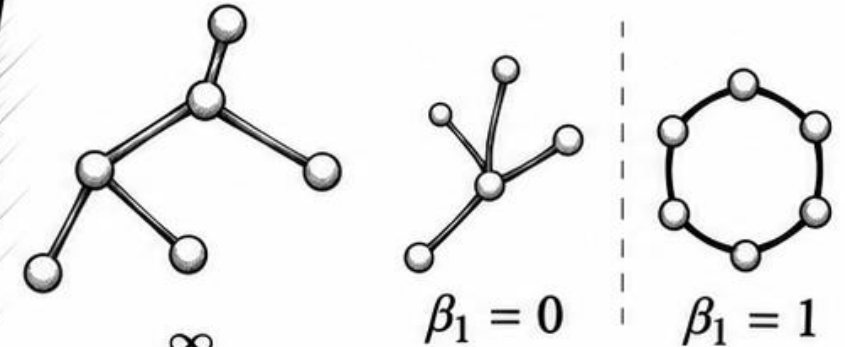
$$E = N - 1$$

Euler characteristic

$$\begin{aligned}\chi &= \# \text{ of edges} - \# \text{ of nodes} \\ &= E - N \\ &= (N - 1) - N \\ &= -1\end{aligned}$$


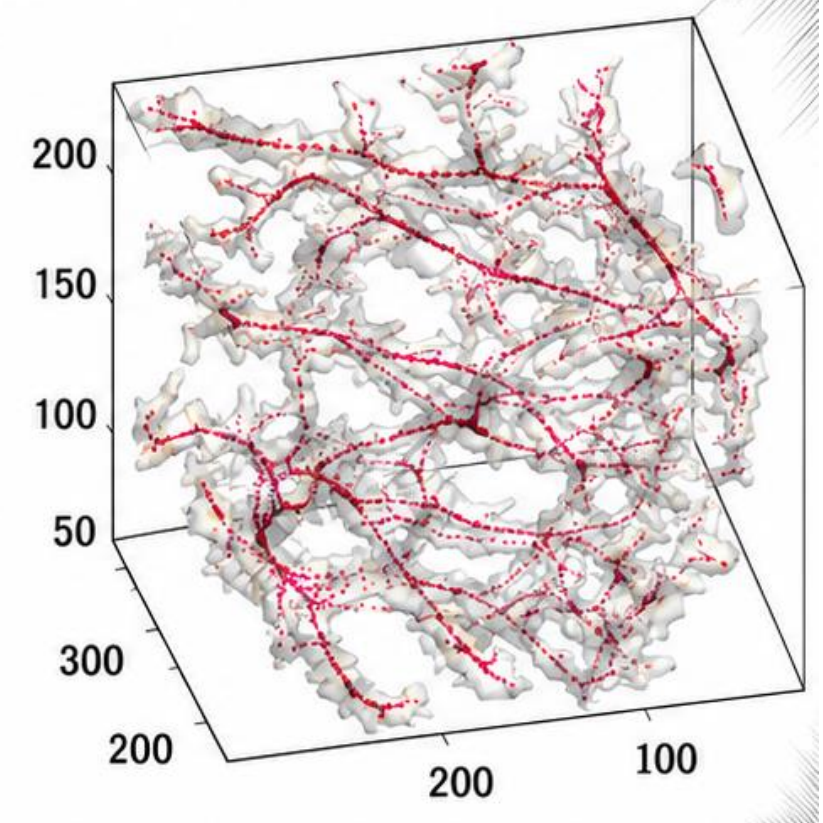
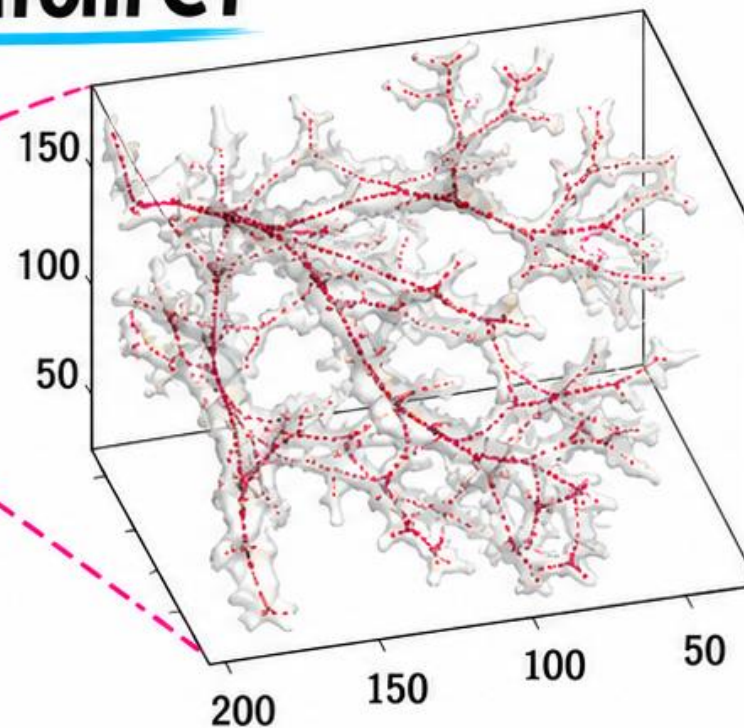
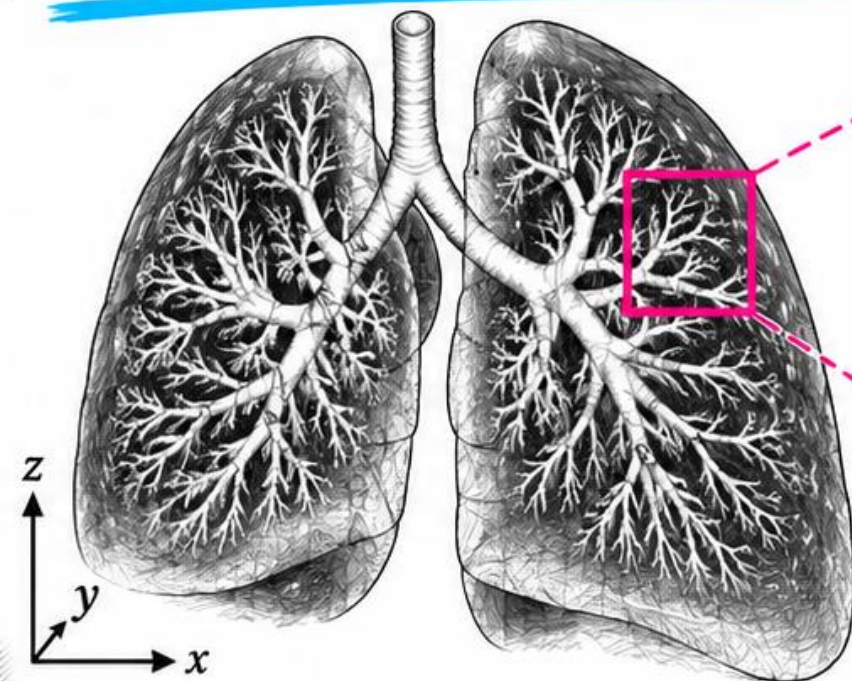
$$\chi = 5 - 6 = -1$$

Betti numbers

$$\beta_0 = 1, \beta_1 = 0$$


$$\begin{aligned}\chi &= \sum_{i=0}^{\infty} (-1)^i \beta_i \\ &= \beta_0 - \beta_1 + \beta_2 - \beta_3 + \dots\end{aligned}$$

Lung blood vessel trees from CT



Chung et al. 2018 EMBC

Chung et al. 2019, Mathematics of Shapes and

Binary images to surface mesh

Marching cubes algorithm

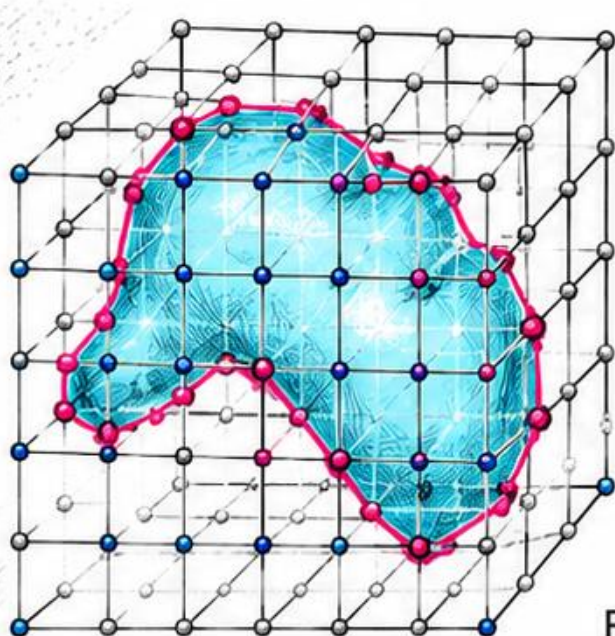
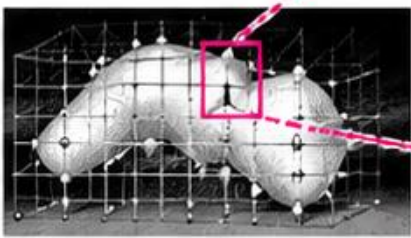


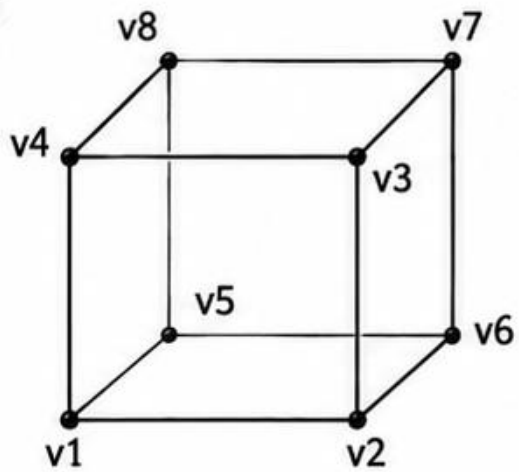
Image intensity values are classified into two parts: in & out

- inside = 1
- outside = 0

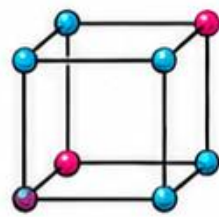


Extract the boundary (iso-surface) from a binary segmentation.

Within each cube

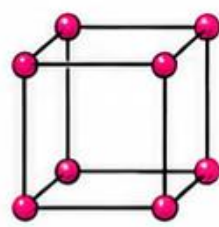


threshold = 9



11110100

threshold = 7



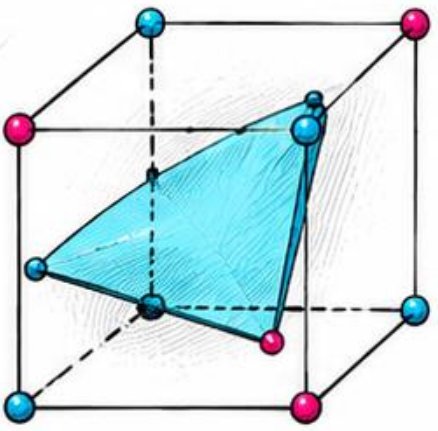
00110000

Index (8-bit):

v1	v2	v3	v4	v5	v6	v7	v8
----	----	----	----	----	----	----	----

Each voxel vertex has a binary label (inside=1, outside=0) based on a threshold.

Interpolate edge intersections

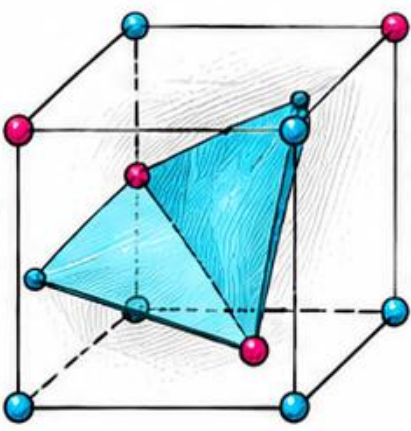


$$p = (1 - t)p_0 + tp_1$$
$$t = \frac{I_0 - \tau}{I_0 - I_1}$$

I_0, I_1 : intensity values
 τ : threshold

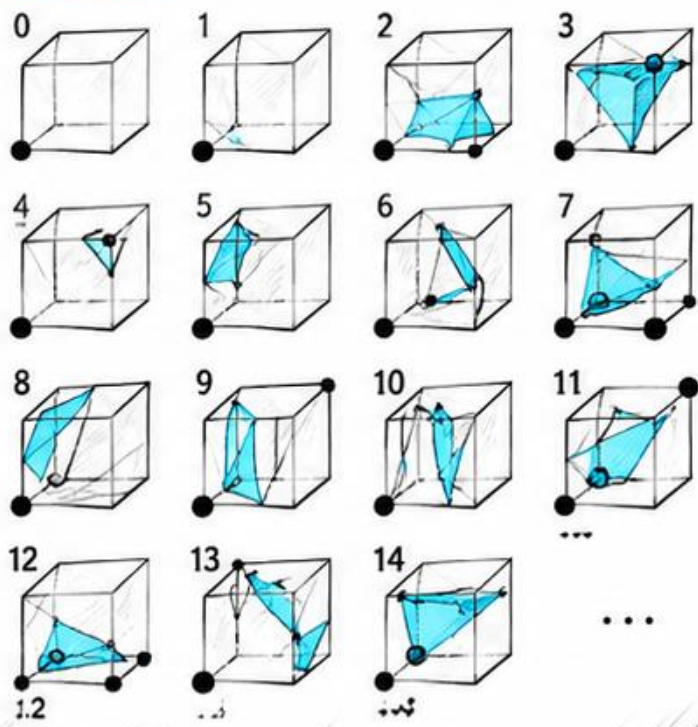
Linearly interpolate the intersection points on edges where the label changes (0 ↔ 1).

Generate triangles



Connect the interpolated points according to the case table to form triangles.

256 cases → 15 unique cases



All 256 cases can be derived from 15 base cases due to symmetries.

Marching cubes in MATLAB

FV =

`isosurface(X,Y,Z,V,ISOVALUE)`

computes isosurface geometry for data V at isosurface value ISOVALUE.

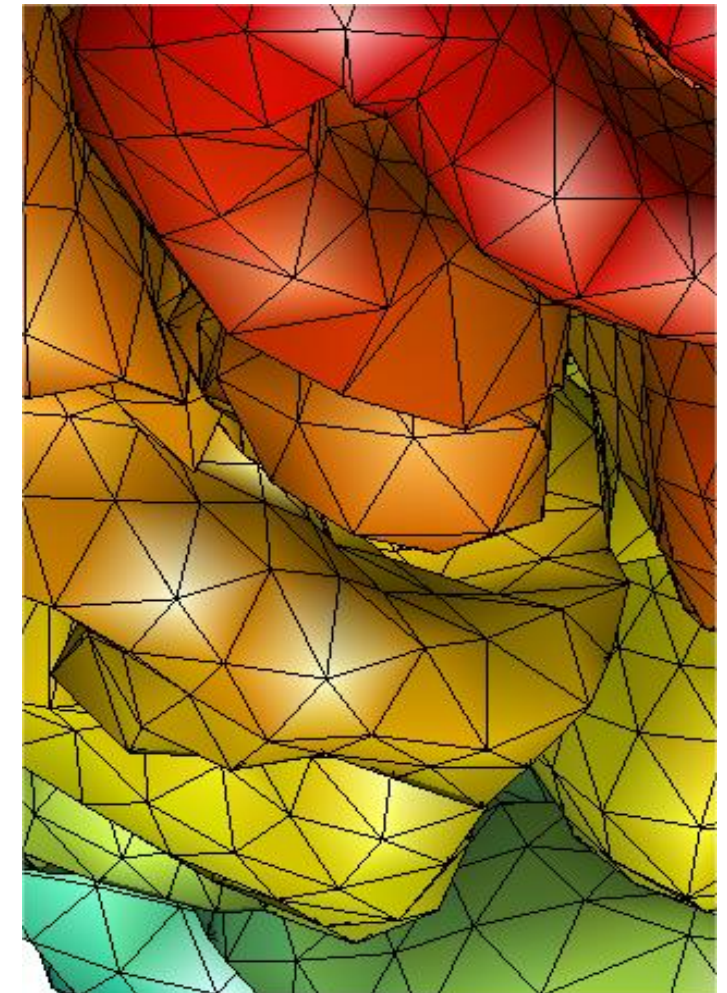
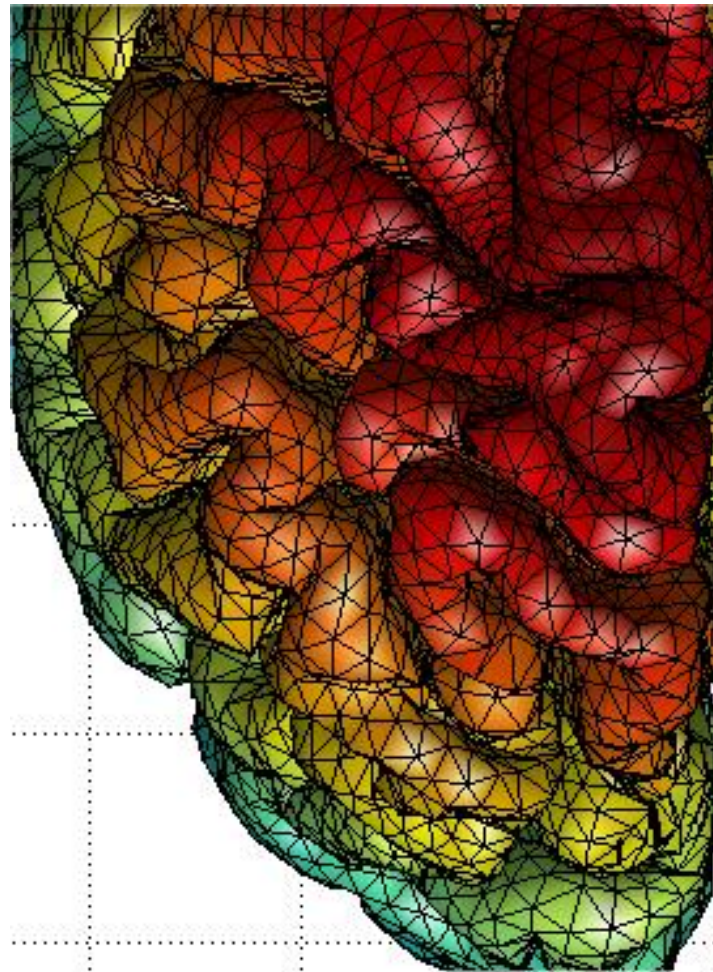
Optional: arrays (X,Y,Z) specify the coordinates at which the data V is given. The struct FV contains the faces and vertices of the isosurface and can be passed directly to the PATCH command.

It is based on simpler version of marching cubes.

Output of isosurface.m

Triangle mesh data structure

Basis of most surface rendering tools for 3D computer games:
as 3D Max Studio, Maya



Data structure for triangle mesh

```
>>surf =  
    vertices: [1282x3 double]  
    faces: [2560x3 double]
```

structured array

```
>>surf.faces
```

```
ans =
```

1	2	3
1	4	2
1	3	5
...		

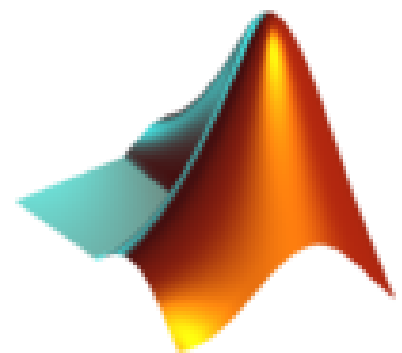
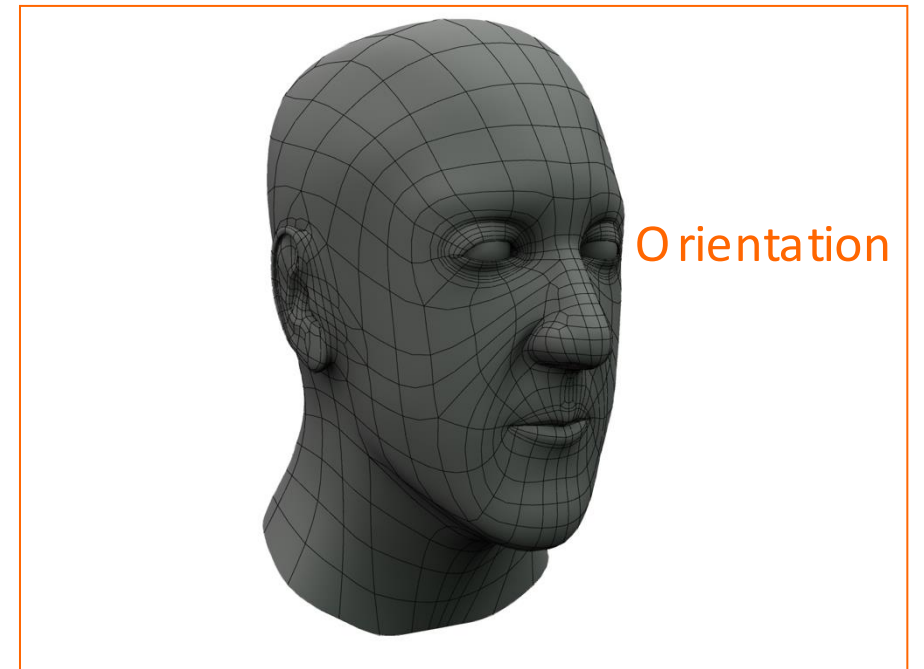
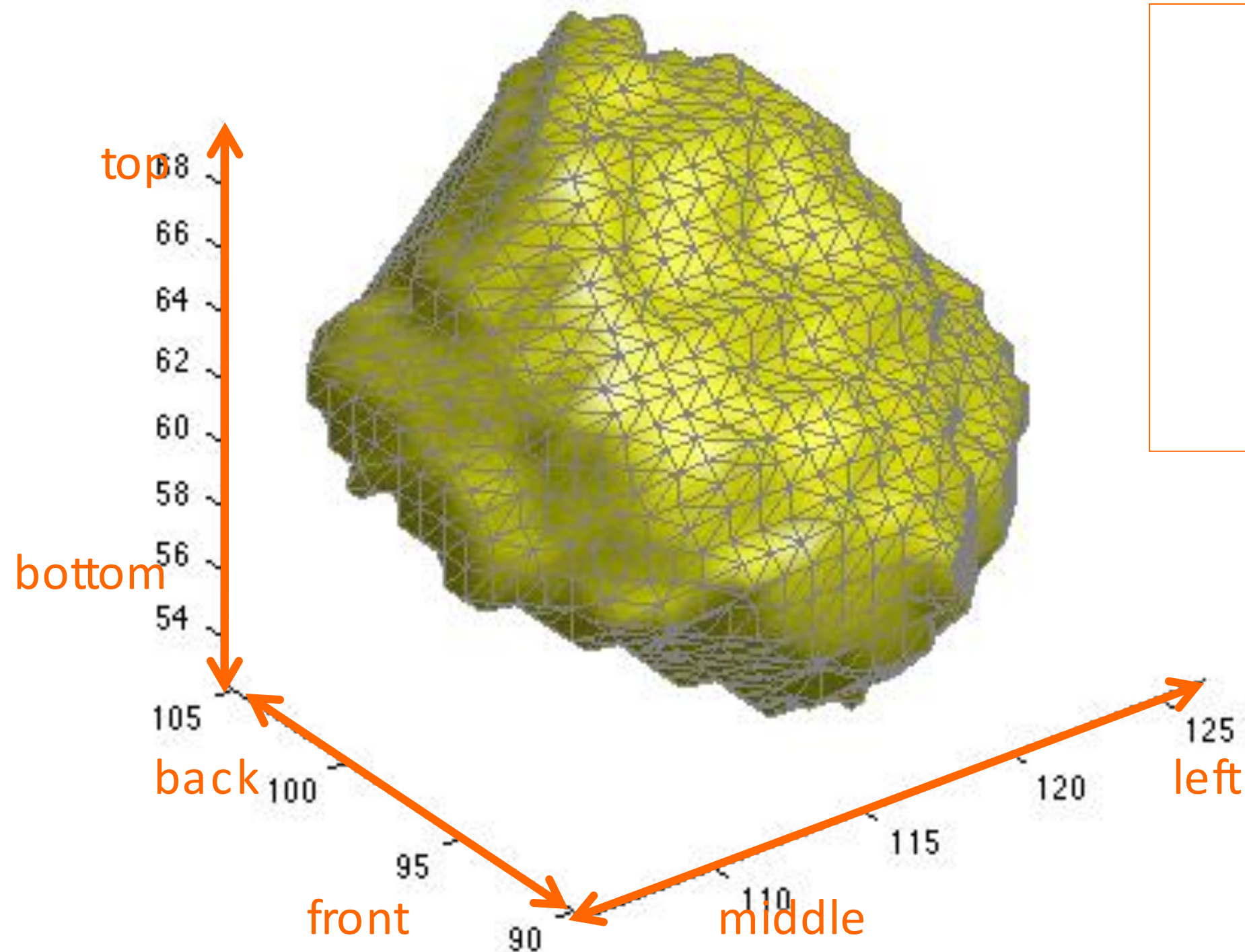
```
>> surf.vertices
```

vertex coordinates

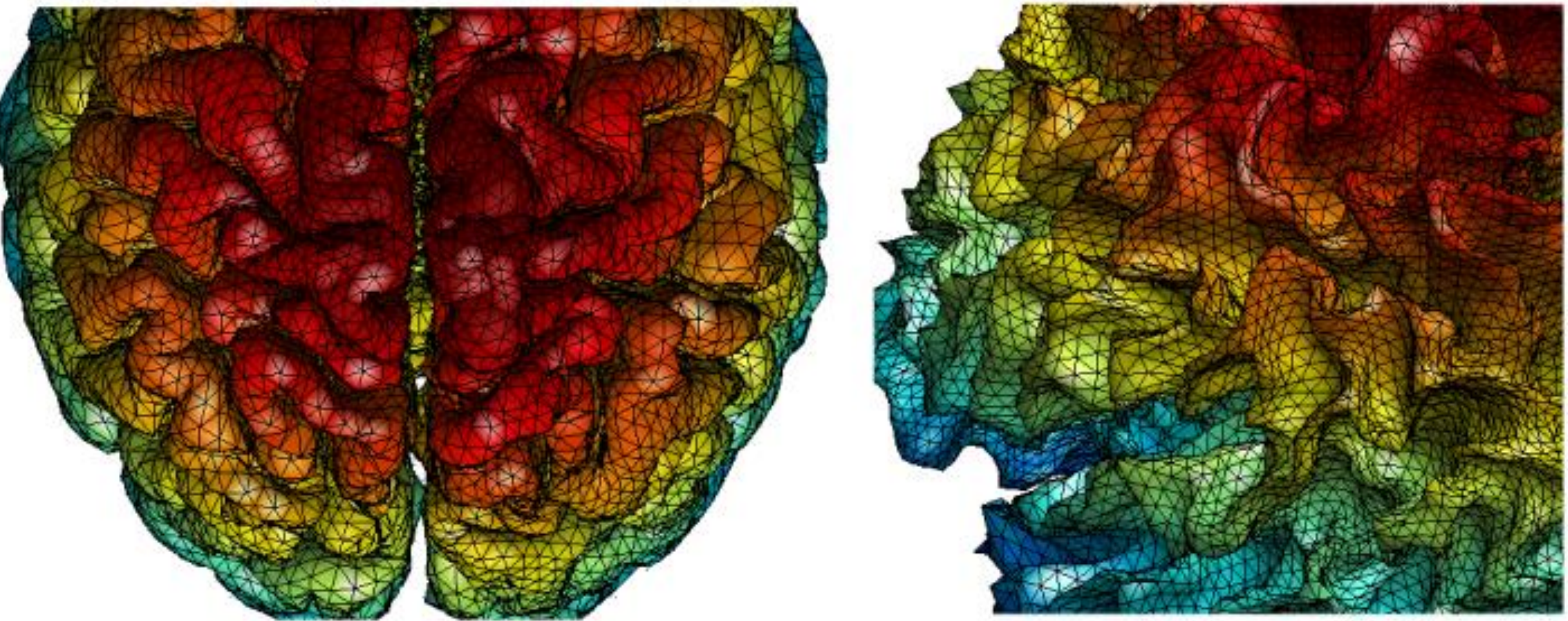
```
ans =
```

75.0000	93.0000	51.5050
74.5050	93.0000	52.0000
75.0000	92.5050	52.0000
...		

2D surface model of left amygdala using marching cubes algorithm



Matlab
demo



Euler characteristic for a surface mesh

How to check the topologically defect of a surface?

$N - E + F = 2$ for a surface topologically equivalent to a sphere.

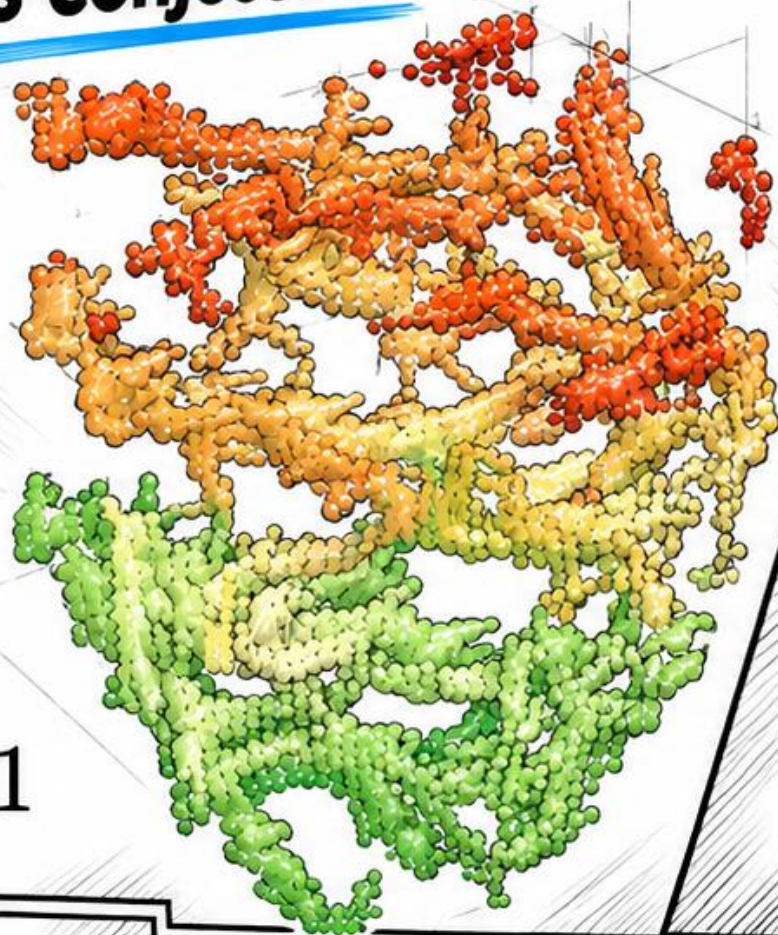
For each triangle, there are three edges. Since two adjacent triangles share the same edge, the total number of edges is

$E = 3F/2$. Hence, we have $F = 2N - 4$ for a closed surface.

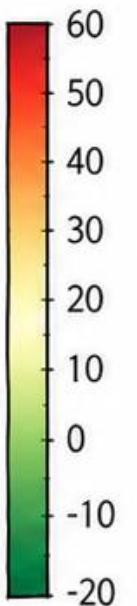
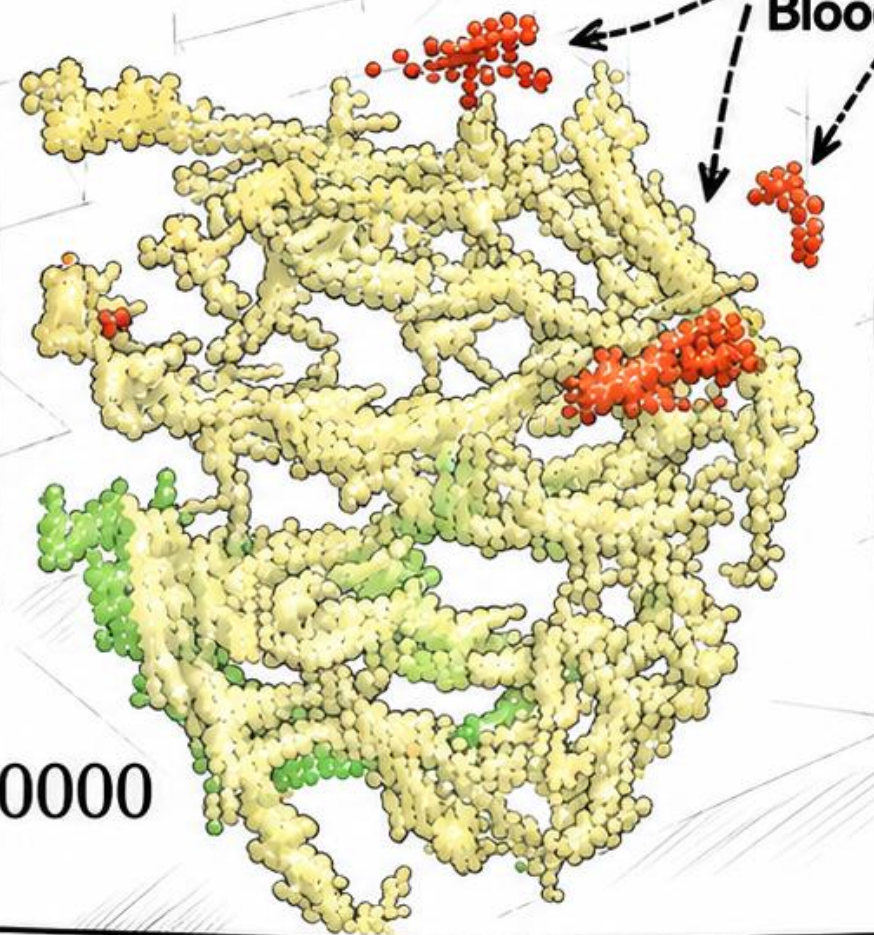
Advanced topology problems

Hot spots conjecture

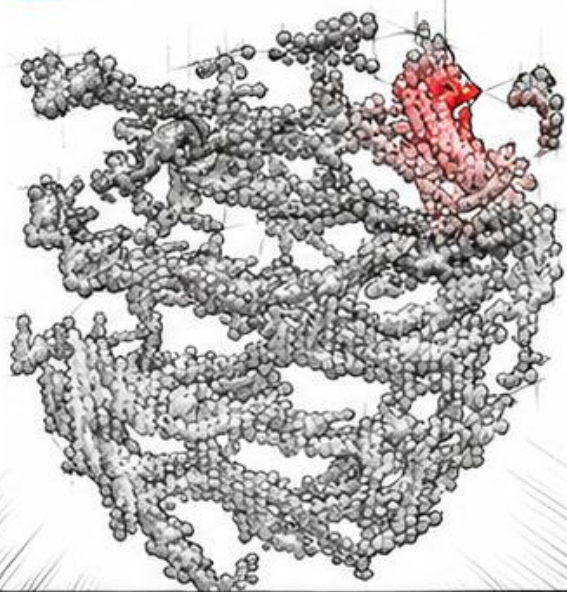
$\sigma = 1$



$\sigma = 10000$



Diffusion on a vascular network



Decomposition

$$K_{\sigma} * f(p) =$$

$$\frac{\int_{\mathcal{M}} f(p) d\mu(p)}{\mu(\mathcal{M})} + f_1 e^{-\lambda_1 \sigma} \psi_1(p) + R(\sigma, p)$$

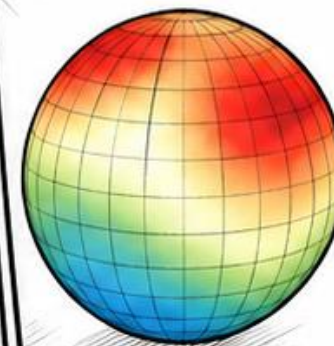
Mean signal

Topology term

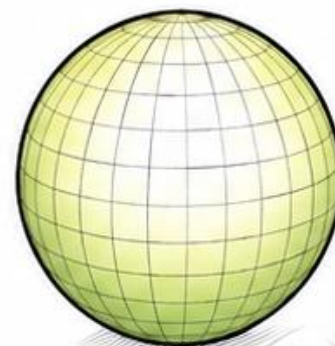
Remainder

Local hot spots (topology-aware)

$\sigma = 1$



$\sigma = 10000$



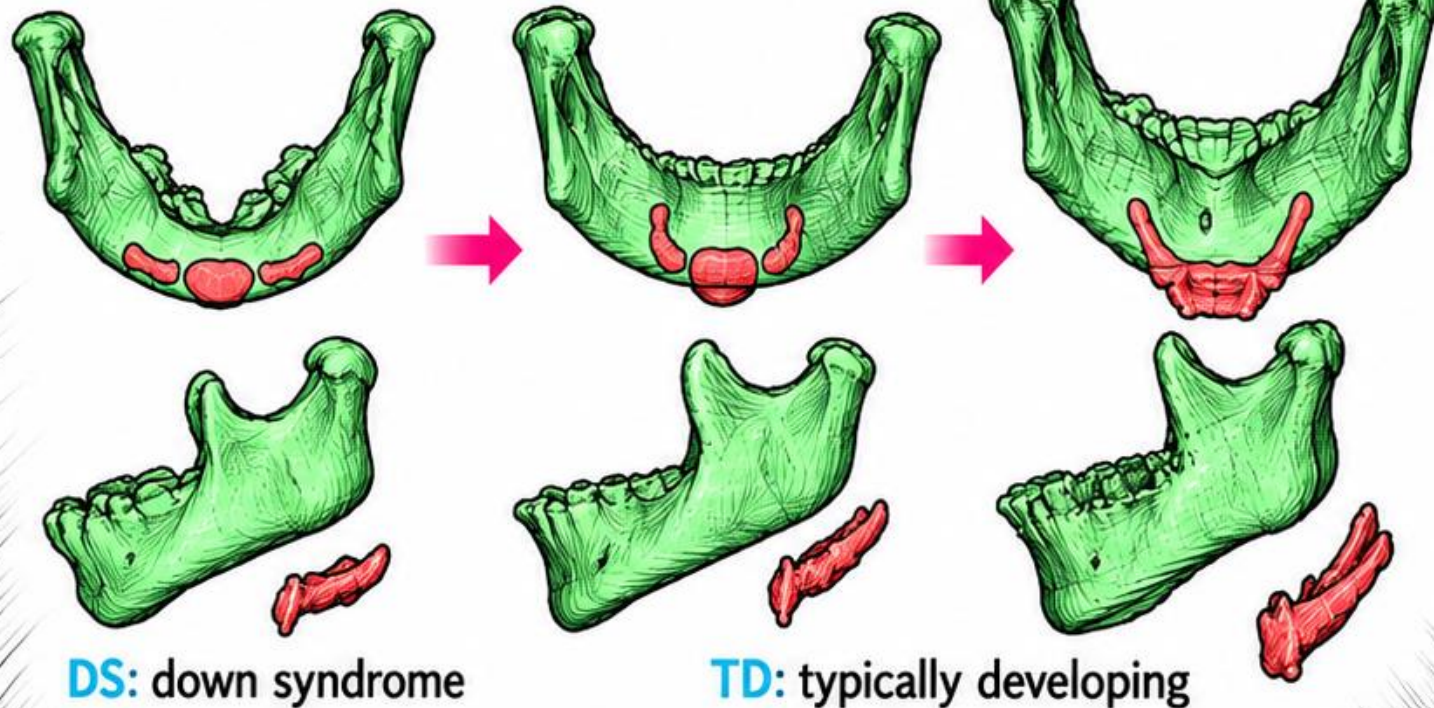
Diffusion smooths out, leaving global structure

Topology changing bone fusion

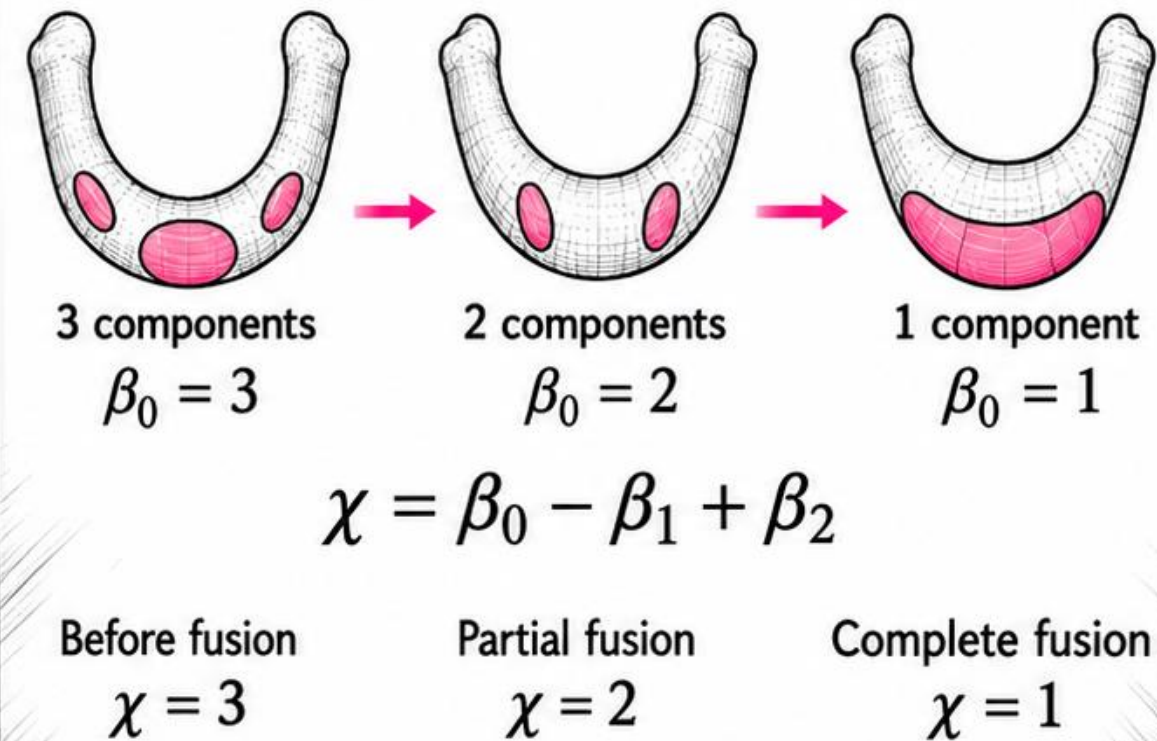
DS; 10 yrs, 6 mo.

TD; 10 yrs, 11 mo.

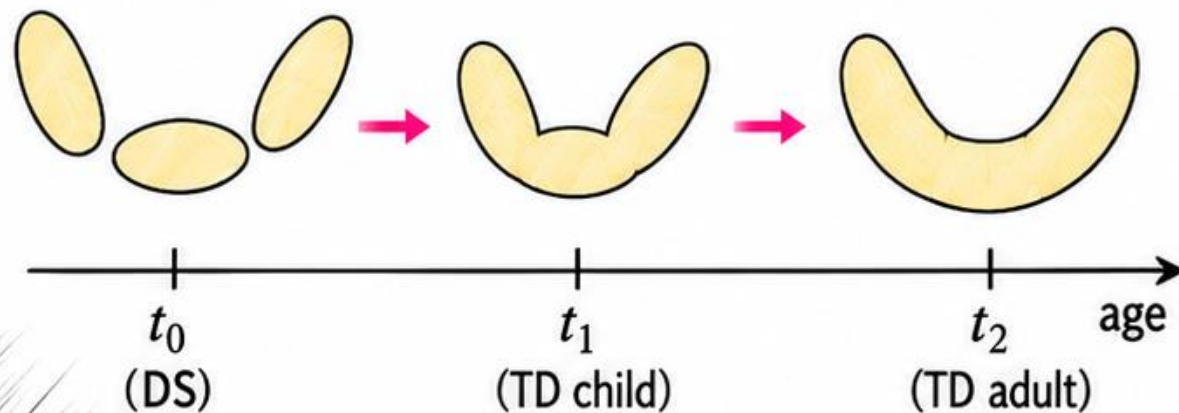
TD; 44 yrs, 1 mo.



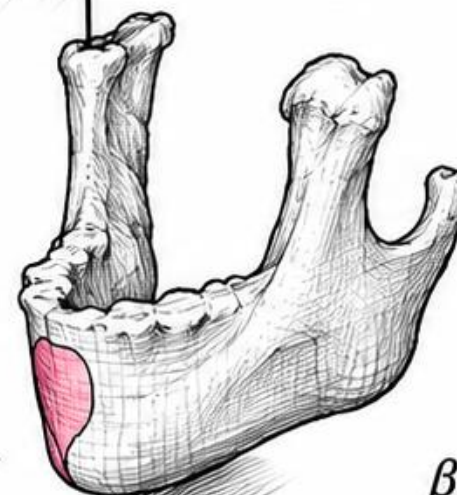
Topological change by bone fusion



Can we build a longitudinal growth model for topologically changing structures?



Modeling framework



$M(t)$: mandibular shape

$T(t)$: topology

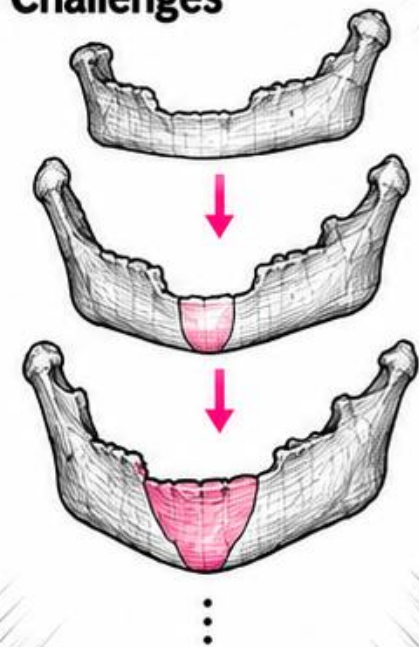
$T(t) = \{\beta_0(t), \beta_1(t), \beta_2(t)\}$

Growth model

$$\frac{dM}{dt} = f(M, t)$$

$\beta_i(t)$ change by fusion events
(topology-preserving between events)

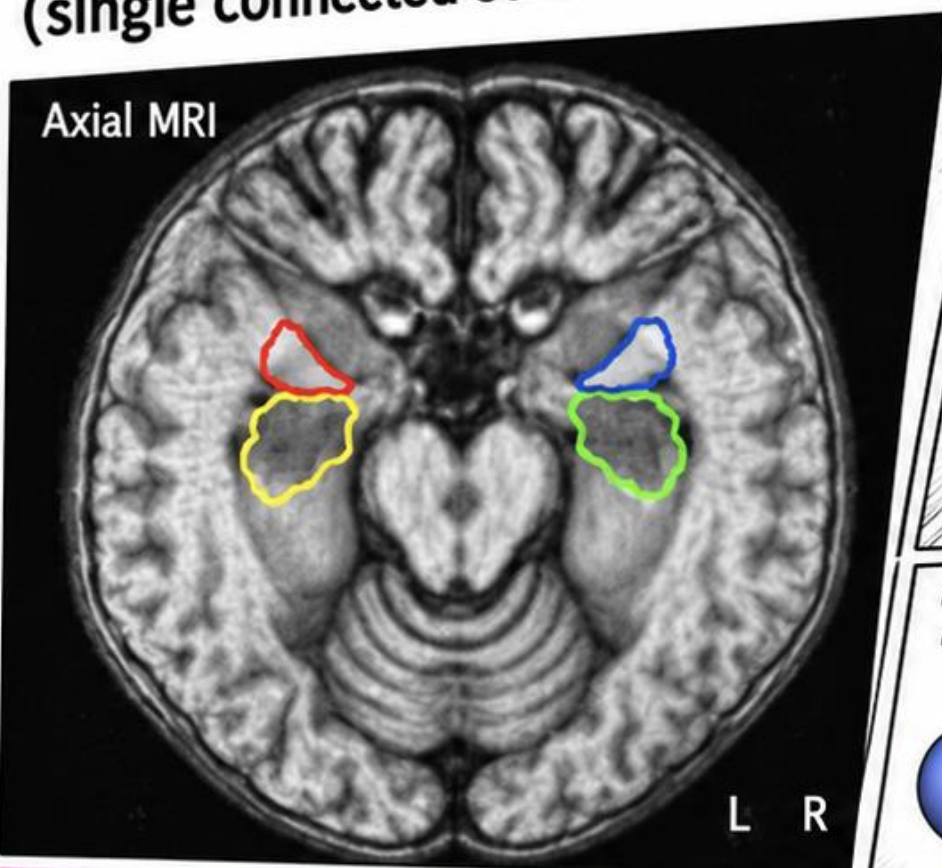
Challenges



Project: This is not trivial modeling task.

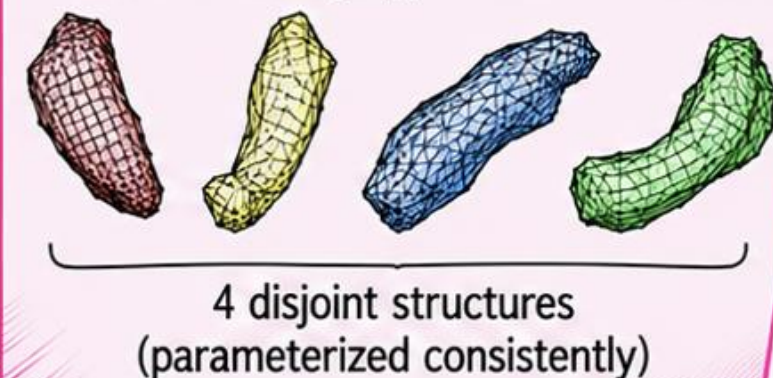
Spherical harmonics (SPHARM)

parameterize with respect to a sphere
(single connected structure)



Problem

How to represent 4 disjoint structures using Hyper-SPHARM basis?



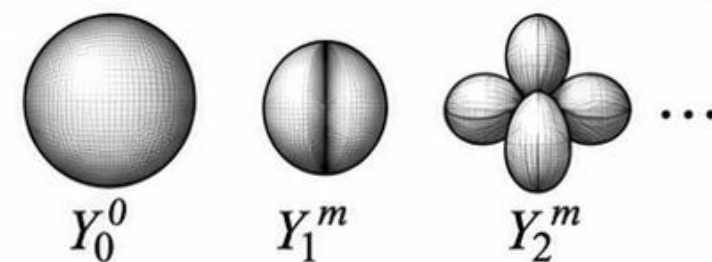
1. SPHARM parameterization

Map a single structure to a sphere and represent the surface using spherical harmonics.

Unit sphere
(parameter domain)

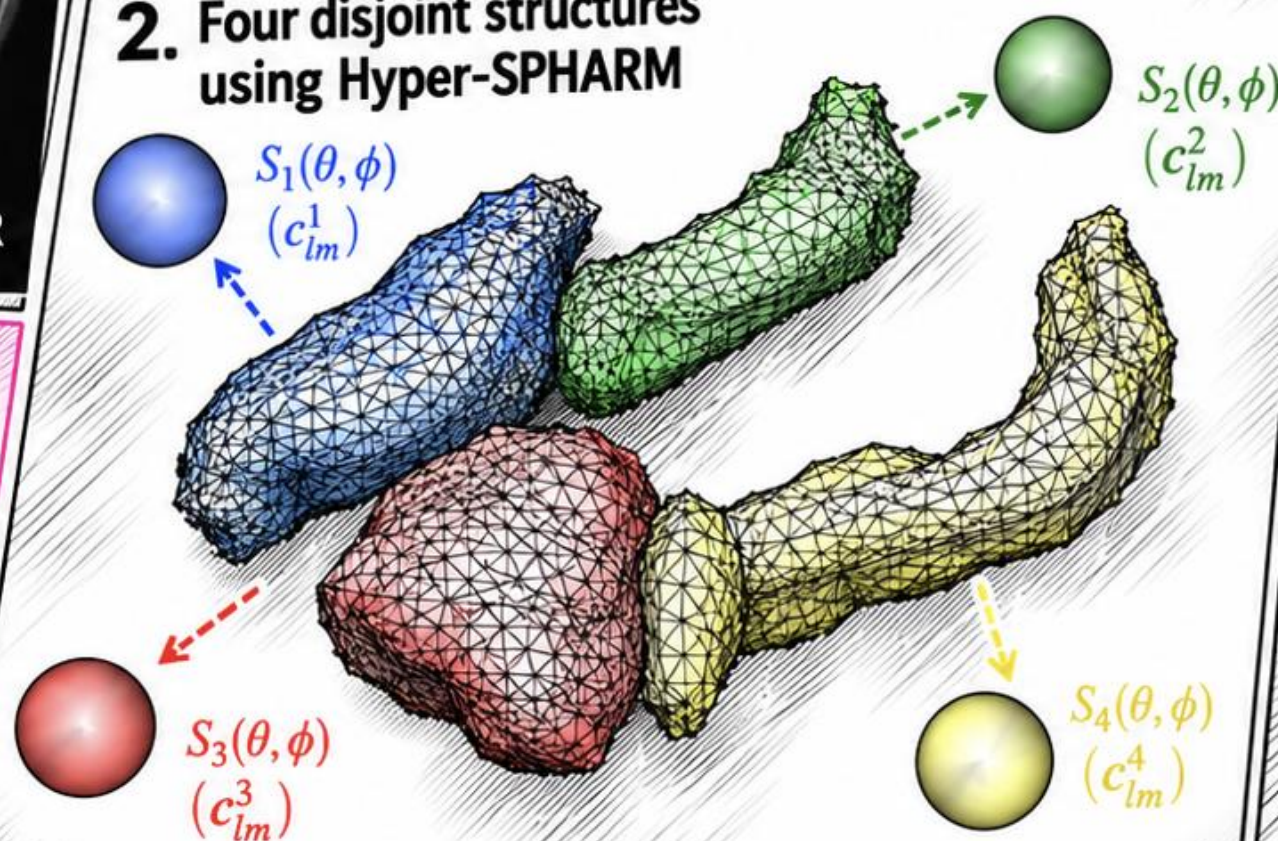
$$x(\theta, \phi) = \sum_{l=0}^L \sum_{m=-l}^l c_{lm} Y_{lm}(\theta, \phi)$$

Spherical harmonics basis



$$x(\theta, \phi) = \sum_{l=0}^L \sum_{m=-l}^l \underbrace{c_{lm}}_{\substack{\text{SPHARM} \\ \text{coefficients} \\ \text{(shape descriptor)}}} Y_{lm}(\theta, \phi)$$

2. Four disjoint structures using Hyper-SPHARM



Hyper-SPHARM representation

$$C = \begin{bmatrix} c^1 \\ c^2 \\ c^3 \\ c^4 \end{bmatrix} \left. \vphantom{\begin{bmatrix} c^1 \\ c^2 \\ c^3 \\ c^4 \end{bmatrix}} \right\} \text{Joint shape representation}$$

$$S_i(\theta, \phi) = \sum_{l=0}^L \sum_{m=-l}^l c_{lm}^i Y_{lm}(\theta, \phi)$$

$i = 1, 2, 3, 4$

Unit Outcome

Able to explain Euler characteristics (EC) and Betti numbers.

Can compute EC and Betti numbers.